AUTOMORPHIC FORMS ON THE HEISENBERG GROUP

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Abstract. Let H be the $(2n+1)$-dimensional real Heisenberg group and Γ its subgroup of integer points. The right regular representation of H on $L^2(\Gamma\backslash H)$ is the simplest nonabelian instance of the representation-theoretic approach to automorphic forms, and, unlike the reductive case, it can be decomposed completely and explicitly. We give a self-contained account. For $m \neq 0$, the part of $L^2(\Gamma\backslash H)$ on which the center acts by $e^{2\pi i m z}$ is a direct sum of $|m|^n$ copies of the Schrödinger representation, and the intertwining operators are theta series; for $m = 1$ the intertwiner is the Weil–Brezin map. We then derive classical consequences: the quasi-periodicity of theta functions is the Heisenberg group law, Poisson summation and the transformation law $\vartheta(-1/\tau) = \sqrt{-i\tau}\,\vartheta(\tau)$ follow from multiplicity one, and the dimension of the space of theta functions of level m is a multiplicity. Finally, we explain the number $|m|^n$ in two further ways: as the dimension of an irreducible representation of a finite Heisenberg group acting on the multiplicity space, and adelically, where every Schrödinger representation occurs exactly once and the factor $|m|^n$ comes from the finite places.

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1. Introduction

A recurring theme in modern number theory is that automorphic forms are best understood as vectors in a representation. If G is a Lie group and $\Gamma \subset G$ a lattice, then G acts on $L^2(\Gamma\backslash G)$ by right translation, and the classical automorphic forms on Γ (modular forms, Maass forms, and their relatives) appear as particularly nice vectors in the irreducible subrepresentations. Decomposing $L^2(\Gamma\backslash G)$ into irreducibles is then the spectral theory of automorphic forms. For $G = SL_2(\mathbb{R})$ and $\Gamma = SL_2(\mathbb{Z})$ this is already a deep problem: the quotient is not compact, so there is continuous spectrum; the Maass cusp forms in the discrete spectrum cannot in general be written down; and the distribution of the spectrum is the subject of open conjectures.

This paper treats a case in which every one of these questions has a complete and explicit answer. Let $H = H_n(\mathbb{R})$ be the Heisenberg group of dimension $2n+1$, and let $\Gamma = H_n(\mathbb{Z})$ be its integer points. The quotient $\Gamma\backslash H$ is a compact manifold, a circle bundle over the torus $\mathbb{T}^{2n}$, and the irreducible unitary representations of H are classified by the Stone–von Neumann theorem. Our main result is the following; precise definitions are in Sections 3–5.

Main Theorem. *Under the right regular representation of H,*

$$L^2(\Gamma\backslash H) \;\cong\; \widehat{\bigoplus}_{(a,b)\in\mathbb{Z}^{2n}} \chi_{a,b} \;\oplus\; \widehat{\bigoplus}_{m\in\mathbb{Z}\setminus\{0\}} \rho_m^{\oplus |m|^n},$$

where $\chi_{a,b}$ is the one-dimensional character $(x,y,z)\mapsto e^{2\pi i(a\cdot x + b\cdot y)}$ and ρ_m is the Schrödinger representation with central character $z\mapsto e^{2\pi i m z}$. The isomorphism on the ρ_m-isotypic part is given by explicit theta series (Theorem 9.2); for $m=1$ it is the Weil–Brezin map

$$(Wf)(x,y,z) = e^{2\pi i z} \sum_{k\in\mathbb{Z}^n} f(x+k)\, e^{2\pi i k\cdot y}.$$

In particular, the space of functions on $\Gamma\backslash H$ on which the center acts by $e^{2\pi i z}$ is a single copy of the Schrödinger representation, and for general m the multiplicity $|m|^n$ is an honest invariant: it is the dimension of $\mathrm{Hom}_H(\rho_m, L^2(\Gamma\backslash H))$ (Corollary 9.5).

The name "automorphic forms on the Heisenberg group" is justified by what happens when one feeds Gaussians into these intertwiners: one obtains exactly the classical Jacobi and Riemann theta functions (Section 10). The representation theory then explains three classical facts at once.

(1) The quasi-periodicity $\vartheta(w+\tau) = e^{-\pi i\tau - 2\pi i w}\vartheta(w)$ is the Heisenberg group law (Proposition 10.2).
(2) Poisson summation, and with it the transformation law $\vartheta(0,-1/\tau) = \sqrt{-i\tau}\,\vartheta(0,\tau)$, follows from the fact that ρ_1 occurs with multiplicity one (Theorem 10.4).
(3) The space of theta functions of level m has dimension m because each of the m copies of ρ_m contains a one-dimensional space of Gaussians (Theorem 10.8).

Outline. Section 2 collects the background on unitary representations: Schur's lemma, central characters, and multiplicity as a dimension. Sections 3 and 4 introduce the Heisenberg group, the lattice Γ, the compact quotient $\Gamma\backslash H$, and the right regular representation. Section 5 constructs the Schrödinger representations, proves

that they are irreducible, and states the Stone–von Neumann theorem. Sections 6–9 prove the Main Theorem in four steps: we split by central character (Section 6), treat the abelian part (Section 7), identify the remaining pieces as Hilbert spaces by a Fourier expansion (Section 8), and identify the group action through the theta map (Section 9). Section 9 also shows that a finite Heisenberg group acts irreducibly on the multiplicity space. Section 10 derives the consequences for theta functions, Section 11 explains how $\Gamma\backslash H$ arises from the adelic quotient $H(\mathbb{Q})\backslash H(\mathbb{A})$, and Section 12 indicates where the subject goes next.

The decomposition goes back to Weil, Cartier, Brezin, Auslander and Tolimieri [Wei64, Car66, Bre70, AT75, Aus77], and from a more geometric point of view to Mumford [Mum91]; the proof here is closest in spirit to the survey of Thangavelu [Tha09]. All proofs are complete except for two classification theorems, which we state with references: the Stone–von Neumann theorem (Theorem 5.4), used only in Corollary 9.6, and Weil's adelic version of it (Theorem 11.5), used only in Section 11.

Conventions. Throughout, $n \geq 1$ is fixed, $x \cdot y$ is the standard inner product on $\mathbb{R}^n$, and

$$\mathbf{e}(t) = e^{2\pi i t} \qquad (t \in \mathbb{R}).$$

The identity element of any group is written 1; the bold letter $\mathbf{e}$ always denotes the function above. Fourier series on $\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$ are written $\varphi(y) = \sum_{k\in\mathbb{Z}^n} c_k\, \mathbf{e}(k \cdot y)$ with $c_k = \int_{[0,1)^n} \varphi(y)\mathbf{e}(-k\cdot y)\, dy$, and the Fourier transform on $\mathbb{R}^n$ is $\mathfrak{F}f(\xi) = \hat{f}(\xi) = \int_{\mathbb{R}^n} f(t)\mathbf{e}(-t\cdot\xi)\, dt$. We write $\mathcal{S}(\mathbb{R}^n)$ for the Schwartz space and $\mathsf{T}_s f(t) = f(t+s)$ for translation. Hilbert spaces are complex, inner products are linear in the first variable, and $\widehat{\bigoplus}$ denotes a Hilbert (completed orthogonal) direct sum.

2. Background on unitary representations

We recall the small amount of general representation theory used below. Standard references are [Fol16, Ch. 3] and [Fol89, Ch. 1].

Definition 2.1. A *unitary representation* of a topological group G on a Hilbert space V is a homomorphism $\pi\colon G \to U(V)$ into the unitary group of V that is *strongly continuous*: for each $v \in V$ the map $g \mapsto \pi(g)v$ is continuous from G to V.

A closed subspace $W \subseteq V$ is *invariant* if $\pi(g)W \subseteq W$ for all g; then π restricts to a *subrepresentation* on W. The representation is *irreducible* if $V \neq 0$ and its only closed invariant subspaces are 0 and V. An *intertwiner* from (π, V) to (σ, V') is a bounded linear map $T\colon V \to V'$ with $T\pi(g) = \sigma(g)T$ for all g; these form a vector space $\mathrm{Hom}_G(\pi,\sigma)$, and we write $\mathrm{End}_G(\pi) = \mathrm{Hom}_G(\pi,\pi)$. Two representations are *equivalent* if there is a unitary intertwiner between them.

Lemma 2.2. *If W is an invariant subspace of a unitary representation (π, V), so is $W^\perp$. If $T \in \mathrm{Hom}_G(\pi,\sigma)$, then $T^* \in \mathrm{Hom}_G(\sigma,\pi)$.*

Proof. If $v \in W^\perp$ and $w \in W$, then $\langle \pi(g)v, w\rangle = \langle v, \pi(g)^{-1}w\rangle = 0$ because $\pi(g)^{-1}w = \pi(g^{-1})w \in W$. For the second claim, take adjoints in $T\pi(g^{-1}) = \sigma(g^{-1})T$ and use $\pi(g^{-1})^* = \pi(g)$ and $\sigma(g^{-1})^* = \sigma(g)$. □

Lemma 2.3 (Schur's lemma)**.** *A unitary representation (π, V) with $V \neq 0$ is irreducible if and only if* $\mathrm{End}_G(\pi) = \mathbb{C}\cdot\mathrm{Id}$.

Proof. If W is a closed invariant subspace, the orthogonal projection P onto W commutes with every $\pi(g)$ by Lemma 2.2, so $\mathrm{End}_G(\pi) = \mathbb{C}$ forces $P \in \{0, \mathrm{Id}\}$. Conversely, suppose π is irreducible and $T \in \mathrm{End}_G(\pi)$. By Lemma 2.2, $T^* \in \mathrm{End}_G(\pi)$, so $T = A + iB$ with $A = \frac{1}{2}(T + T^*)$ and $B = \frac{1}{2i}(T - T^*)$ self-adjoint elements of $\mathrm{End}_G(\pi)$. If A is self-adjoint and commutes with $\pi(G)$, then so do all its spectral projections; their ranges are closed invariant subspaces, hence 0 or V, so the spectral measure of A is a point mass and A is a scalar. The same holds for B, hence for T. □

Corollary 2.4 (Central characters)**.** *Let Z be the center of G. If π is irreducible, there is a continuous homomorphism $\psi\colon Z \to U(1)$ with $\pi(z) = \psi(z)\,\mathrm{Id}$ for all $z \in Z$.*

Proof. Each $\pi(z)$ commutes with $\pi(G)$, so it is a scalar by Lemma 2.3; the scalar has modulus one because $\pi(z)$ is unitary, and it depends continuously on z by strong continuity. □

We call ψ the *central character* of π. More generally, we say an arbitrary representation *has central character* ψ if $\pi(z) = \psi(z)\mathrm{Id}$ for all $z \in Z$.

The following lemma makes precise the idea that a multiplicity is a dimension.

Lemma 2.5 (Multiplicity as a dimension)**.** *Let σ be irreducible, and let $V = \widehat{\bigoplus}_{i=1}^{N} V_i$ be an orthogonal direct sum of invariant subspaces of a unitary representation (π, V), with $N < \infty$. Suppose there are unitary intertwiners $T_i\colon \sigma \xrightarrow{\sim} \pi|_{V_i}$. Then $\mathrm{Hom}_G(\sigma, \pi)$ has basis $T_1, \dots, T_N$ (regarded as maps into V), and in particular $\dim \mathrm{Hom}_G(\sigma, \pi) = N$.*

Proof. The T_i are linearly independent because their images are mutually orthogonal and nonzero. Let $S \in \mathrm{Hom}_G(\sigma, \pi)$ and let P_i be the orthogonal projection onto V_i. Since V_i and its complement are invariant, P_i is an intertwiner, so $T_i^{-1} P_i S \in \mathrm{End}_G(\sigma)$ is a scalar λ_i by Lemma 2.3. Hence $P_i S = \lambda_i T_i$ and $S = \sum_i P_i S = \sum_i \lambda_i T_i$. □

3. The Heisenberg group

3.1. Definition and first properties.

Definition 3.1. The *Heisenberg group* $H = H_n(\mathbb{R})$ is the set $\mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}$ with multiplication

$$(x, y, z)(a, b, c) = (x + a,\ y + b,\ z + c + x \cdot b). \tag{3.1}$$

The identity is $(0, 0, 0)$, and one checks directly that

$$(x, y, z)^{-1} = (-x,\ -y,\ -z + x \cdot y),$$

since $(x, y, z)(-x, -y, -z + x \cdot y) = (0, 0, z - z + x \cdot y - x \cdot y)$. Associativity is also a direct computation: in both $\big((x, y, z)(a, b, c)\big)(p, q, r)$ and $(x, y, z)\big((a, b, c)(p, q, r)\big)$ the first two coordinates are $x + a + p$ and $y + b + q$, and the third coordinates are

$$z + c + x \cdot b + r + (x + a) \cdot q \qquad \text{and} \qquad z + (c + r + a \cdot q) + x \cdot (b + q),$$

which agree. A more conceptual proof is that (x, y, z) corresponds to the unipotent matrix

$$\begin{pmatrix} 1 & x & z \\ 0 & I_n & y \\ 0 & 0 & 1 \end{pmatrix} \in GL_{n+2}(\mathbb{R}), \tag{3.2}$$

with x a row vector and y a column vector, and that matrix multiplication reproduces (3.1). In particular H is a closed subgroup of $GL_{n+2}(\mathbb{R})$, hence a Lie group; it is diffeomorphic to $\mathbb{R}^{2n+1}$.

Proposition 3.2. *For $g = (x, y, z)$ and $g' = (a, b, c)$ in H,*

$$gg' = g'g \cdot \big(0, 0,\ x \cdot b - a \cdot y\big). \tag{3.3}$$

Consequently the center of H and the commutator subgroup $[H, H]$ both equal

$$Z = \{(0, 0, s) : s \in \mathbb{R}\} \cong \mathbb{R},$$

and $H/Z \cong \mathbb{R}^{2n}$ via $(x, y, z) \mapsto (x, y)$.

Proof. Both gg' and $g'g$ have first two coordinates $(x + a, y + b)$, and their third coordinates differ by $x \cdot b - a \cdot y$; since $(0, 0, s)$ commutes with everything and multiplies on the right by adding s to the third coordinate, (3.3) follows. Thus g' is central if and only if $x \cdot b = a \cdot y$ for all x, y, i.e., $a = b = 0$. The commutators $gg'g^{-1}g'^{-1} = (0, 0, x \cdot b - a \cdot y)$ exhaust Z. □

The bilinear form $\omega\big((x, y), (a, b)\big) = x \cdot b - a \cdot y$ appearing in (3.3) is the standard symplectic form on $\mathbb{R}^{2n}$. So H is a central extension

$$0 \to \mathbb{R} \to H \to \mathbb{R}^{2n} \to 0$$

whose commutator pairing is ω; this is the defining feature of a Heisenberg group.

Remark 3.3 (The symmetric model). Many references, including [Fol89, Tha98], use the group law $(x, y, z)(a, b, c) = (x + a, y + b, z + c + \frac{1}{2}(x \cdot b - y \cdot a))$. The map $(x, y, z) \mapsto (x, y, z - \frac{1}{2}x \cdot y)$ is an isomorphism from our group onto that one, as one checks by expanding both sides. Everything below can be transported along it. We prefer (3.1) because its integer points form a subgroup.

3.2. **Haar measure.** For $g = (a, b, c)$, left translation $L_g(h) = gh$ is the map

$$(x, y, z) \longmapsto (a + x,\ b + y,\ c + z + a \cdot y).$$

This is affine, and its Jacobian matrix is the identity except for the entries a in the z-row and y-columns. Its determinant is 1. Right translation $(x, y, z) \mapsto (x + a, y + b, z + c + x \cdot b)$ likewise has Jacobian determinant 1. Hence:

Proposition 3.4. *Lebesgue measure $dh = dx\,dy\,dz$ on $\mathbb{R}^{2n+1}$ is a left and right Haar measure on H. In particular H is unimodular.*

3.3. **Invariant vector fields.** Let $e_1, \dots, e_n$ be the standard basis of $\mathbb{R}^n$. Right translation by the one-parameter subgroups $s \mapsto (se_j, 0, 0)$, $s \mapsto (0, se_j, 0)$ and $s \mapsto (0, 0, s)$ is

$$(x, y, z)(se_j, 0, 0) = (x + se_j, y, z), \qquad (x, y, z)(0, se_j, 0) = (x, y + se_j, z + sx_j),$$
$$(x, y, z)(0, 0, s) = (x, y, z + s).$$

Differentiating at $s = 0$ gives the left-invariant vector fields

$$X_j = \partial_{x_j}, \qquad Y_j = \partial_{y_j} + x_j\,\partial_z, \qquad Z = \partial_z, \tag{3.4}$$

which satisfy $[X_j, Y_k] = \delta_{jk} Z$ and all other brackets zero. These are the Heisenberg commutation relations.

3.4. **An automorphism.** In the symmetric model of Remark 3.3, every σ in the group $Sp_{2n}(\mathbb{R})$ of linear maps preserving ω gives an automorphism $(v, z) \mapsto (\sigma v, z)$ that fixes the center; transported to our coordinates, the z-coordinate acquires a correction term. We need one such automorphism explicitly, the one coming from $\sigma(x, y) = (-y, x)$:

$$J(x, y, z) = (-y,\ x,\ z - x \cdot y). \tag{3.5}$$

Lemma 3.5. *J is an automorphism of H that fixes Z pointwise, preserves Haar measure, and maps $H_n(\mathbb{Z})$ onto itself. Its inverse is $J^{-1}(x, y, z) = (y, -x, z - x \cdot y)$.*

Proof. For $g = (x, y, z)$, $g' = (x', y', z')$, the first two coordinates of $J(g)J(g')$ and $J(gg')$ are both $(-y - y', x + x')$. Their third coordinates are

$$(z - x \cdot y) + (z' - x' \cdot y') + (-y) \cdot x' \quad \text{and} \quad z + z' + x \cdot y' - (x + x') \cdot (y + y'),$$

which are both equal to $z + z' - x \cdot y - x' \cdot y' - x' \cdot y$. The formula for J^{-1} is checked by composing. The linear part of J on (x, y) is a rotation, and the z-coordinate is shifted by a function of (x, y), so the Jacobian determinant is 1. Clearly J and J^{-1} preserve integrality, and $J(0, 0, s) = (0, 0, s)$. □

4. The lattice and the quotient

4.1. **The lattice.** Let $\Gamma = H_n(\mathbb{Z}) = \{(a, b, c) : a, b \in \mathbb{Z}^n,\ c \in \mathbb{Z}\}$, the integer matrices in the model (3.2). It is a subgroup because $a \cdot b' \in \mathbb{Z}$ for $a, b' \in \mathbb{Z}^n$ and because $(a, b, c)^{-1} = (-a, -b, -c + a \cdot b)$ is again integral. It is discrete in H. We record:

$$\Gamma \cap Z = \{(0, 0, c) : c \in \mathbb{Z}\} \cong \mathbb{Z}, \qquad \text{image of } \Gamma \text{ in } H/Z = \mathbb{Z}^{2n}.$$

Remark 4.1. By (3.3), the commutator subgroup of Γ is $\Gamma \cap Z$, so $\Gamma^{\mathrm{ab}} \cong \mathbb{Z}^{2n}$. Hence $H_1(\Gamma\backslash H; \mathbb{Z}) \cong \mathbb{Z}^{2n}$, and $\Gamma\backslash H$ is not homeomorphic to the torus $\mathbb{T}^{2n+1}$, even though it is a $(2n + 1)$-dimensional compact manifold covered by $\mathbb{R}^{2n+1}$. It is a nontrivial circle bundle over $\mathbb{T}^{2n}$.

4.2. **A fundamental domain.**

Proposition 4.2. *Let $D = [0, 1)^{2n+1} \subset H$. Every $h \in H$ can be written uniquely as $h = \gamma d$ with $\gamma \in \Gamma$ and $d \in D$. Equivalently, H is the disjoint union $\bigsqcup_{\gamma \in \Gamma} \gamma D$.*

Proof. For $\gamma = (a, b, c) \in \Gamma$ and $h = (x, y, z)$ we have

$$\gamma h = (x + a,\ y + b,\ z + c + a \cdot y). \tag{4.1}$$

Existence. Choose $a = -\lfloor x \rfloor$ and $b = -\lfloor y \rfloor$ coordinatewise, so $x + a, y + b \in [0, 1)^n$. Then $a \cdot y$ is a real number, and we choose $c = -\lfloor z + a \cdot y \rfloor$. Now $\gamma h \in D$, so $h = \gamma^{-1}(\gamma h)$.

Uniqueness. Suppose $d = (x, y, z)$ and γd both lie in D. From (4.1), x and $x + a$ both lie in $[0, 1)^n$ with $a \in \mathbb{Z}^n$, so $a = 0$; likewise $b = 0$. Then $\gamma d = (x, y, z + c)$, and $z, z + c \in [0, 1)$ forces $c = 0$. □

Consequently $\Gamma\backslash H$ is compact, being the image of the compact set $\overline{D}$, and D has volume 1. The disjoint union of Proposition 4.2 and the left invariance of dh give the *unfolding formula*

$$\int_H f(h)\,dh = \sum_{\gamma\in\Gamma}\int_{\gamma D} f(h)\,dh = \int_D \sum_{\gamma\in\Gamma} f(\gamma h)\,dh \qquad (f\in L^1(H)). \tag{4.2}$$

Lemma 4.3. *Let $D'\subseteq H$ be measurable with $H=\bigsqcup_{\gamma\in\Gamma}\gamma D'$. If F is measurable, nonnegative or integrable over D, and left Γ-invariant, then $\int_{D'}F=\int_D F$.*

Proof. Write $D'=\bigsqcup_\gamma (D'\cap\gamma D)$. The sets $\gamma^{-1}(D'\cap\gamma D)=\gamma^{-1}D'\cap D$ partition D, because the $\gamma^{-1}D'$ partition H. By left invariance of dh and of F,

$$\int_{D'}F=\sum_\gamma\int_{D'\cap\gamma D}F=\sum_\gamma\int_{\gamma^{-1}D'\cap D}F=\int_D F. \qquad \square$$

4.3. **The right regular representation.** We identify functions on $\Gamma\backslash H$ with left Γ-invariant functions on H and give $\Gamma\backslash H$ the measure induced from D. Thus

$$L^2(\Gamma\backslash H)=\Big\{F\colon H\to\mathbb{C}:\ F(\gamma h)=F(h)\ \forall\gamma\in\Gamma,\quad \|F\|^2=\int_D|F(h)|^2\,dh<\infty\Big\},$$

where F ranges over measurable functions modulo those vanishing almost everywhere. For $g\in H$ define

$$(R(g)F)(h)=F(hg).$$

Proposition 4.4. *R is a unitary representation of H on $L^2(\Gamma\backslash H)$.*

Proof. If F is left Γ-invariant, so is $R(g)F$, since $F(\gamma hg)=F(hg)$. Next, $(R(g)R(g')F)(h)=(R(g')F)(hg)=F(hgg')=(R(gg')F)(h)$, so R is a homomorphism. For unitarity, the substitution $h\mapsto hg^{-1}$, which preserves dh, gives

$$\|R(g)F\|^2=\int_D|F(hg)|^2\,dh=\int_{Dg}|F(h)|^2\,dh.$$

The set Dg is again a fundamental domain, since $\Gamma(Dg)=(\Gamma D)g=H$ and right multiplication by g is a bijection, so $\|R(g)F\|=\|F\|$ by Lemma 4.3. Since $R(g)$ has inverse $R(g^{-1})$, it is unitary.

For strong continuity, let F be continuous on $\Gamma\backslash H$. By compactness F is uniformly continuous, so $\sup_h|F(hg)-F(h)|\to 0$ as $g\to 1$, and hence $\|R(g)F-F\|\to 0$. Continuous functions are dense in $L^2(\Gamma\backslash H)$ and the operators $R(g)$ all have norm one, so an $\varepsilon/3$ argument gives $\|R(g)F-F\|\to 0$ for every F. Continuity at an arbitrary g_0 follows from $R(g)F-R(g_0)F=R(g_0)\big(R(g_0^{-1}g)F-F\big)$. $\square$

Remark 4.5 (Why $\Gamma\backslash H$ and not H/Γ). The notation $\Gamma\backslash H$ denotes the space of right cosets Γh, i.e., the orbits of Γ acting on H by *left* multiplication. The lattice is put on the left so that the right side is free for the group action: functions on $\Gamma\backslash H$ are the left Γ-invariant functions, and H acts on them by right translation, as above. This is the standard convention in the theory of automorphic forms. There the group $G=SL_2(\mathbb{R})$ acts on the upper half plane $\mathfrak{H}=G/SO(2)$ by Möbius transformations on the left, a modular form transforms under $\gamma\in SL_2(\mathbb{Z})$ acting on $\mathfrak{H}$ from the left, and so the natural space is the double coset space $SL_2(\mathbb{Z})\backslash SL_2(\mathbb{R})/SO(2)$: the arithmetic group acts on the left and the compact group on the right. The adelic quotients $G(\mathbb{Q})\backslash G(\mathbb{A})$ of Section 11 follow the same pattern. Much of the literature on nilmanifolds and homogeneous dynamics [Moo65, Ric71, CG90] instead writes

G/Γ, with G acting on the left by $(\lambda(g)F)(x) = F(g^{-1}x)$. The two conventions are equivalent: the map $h \mapsto h^{-1}$ induces a homeomorphism $\Gamma\backslash H \to H/\Gamma$ that carries Haar measure to Haar measure (as H is unimodular) and intertwines right translation with left translation.

Remark 4.6. For any cocompact lattice Γ in a unimodular Lie group G, the right regular representation on $L^2(\Gamma\backslash G)$ is a Hilbert direct sum of irreducible representations, each occurring with finite multiplicity [GGPS69, Ch. 1]. The point of the present paper is that for the Heisenberg group every summand and every multiplicity can be determined explicitly.

5. The Schrödinger representations

5.1. Construction.

Definition 5.1. For $\lambda \in \mathbb{R}\setminus\{0\}$, the *Schrödinger representation* ρ_λ of H on $L^2(\mathbb{R}^n)$ is

$$\big(\rho_\lambda(a,b,c)f\big)(t) = \mathbf{e}\big(\lambda(c + t\cdot b)\big)\, f(t+a). \tag{5.1}$$

For $\lambda \in \mathbb{R}$ write $\psi_\lambda(s) = \mathbf{e}(\lambda s)$ for the corresponding character of $Z \cong \mathbb{R}$.

Proposition 5.2. *ρ_λ is a unitary representation of H with central character ψ_λ. It preserves $\mathcal{S}(\mathbb{R}^n)$, and for $f \in \mathcal{S}(\mathbb{R}^n)$ the derived action of the vector fields (3.4) is*

$$d\rho_\lambda(X_j)f = \partial_{t_j} f, \qquad d\rho_\lambda(Y_j)f = 2\pi i\lambda\, t_j f, \qquad d\rho_\lambda(Z)f = 2\pi i\lambda f, \tag{5.2}$$

where the derivatives $\frac{d}{ds}\rho_\lambda(\exp sV)f|_{s=0}$ exist in the norm of $L^2(\mathbb{R}^n)$.

Proof. Apply $\rho_\lambda(a',b',c')$ and then $\rho_\lambda(a,b,c)$:

$$\rho_\lambda(a,b,c)\rho_\lambda(a',b',c')f(t) = \mathbf{e}\big(\lambda(c + t\cdot b)\big)\,\mathbf{e}\big(\lambda(c' + (t+a)\cdot b')\big)\, f(t+a+a').$$

Since $(a,b,c)(a',b',c') = (a+a', b+b', c+c'+a\cdot b')$, we also have

$$\rho_\lambda\big((a,b,c)(a',b',c')\big)f(t) = \mathbf{e}\big(\lambda(c + c' + a\cdot b' + t\cdot b + t\cdot b')\big)\, f(t+a+a'),$$

and the exponents agree. Each $\rho_\lambda(g)$ is a translation followed by multiplication by a function of modulus one, hence unitary. Strong continuity follows from the continuity of translation in $L^2(\mathbb{R}^n)$ and dominated convergence for the multiplier. The central character is visible from (5.1) with $a = b = 0$. Translations and multiplication by $\mathbf{e}(\lambda t\cdot b)$ preserve $\mathcal{S}(\mathbb{R}^n)$. Finally (5.2) follows by differentiating (5.1) along the three one-parameter subgroups of §3.3; for f Schwartz the difference quotients converge in L^2 by dominated convergence. □

5.2. Irreducibility.

Proposition 5.3. *For every $\lambda \neq 0$, the representation ρ_λ is irreducible.*

Proof. By Lemma 2.3 it suffices to show that every bounded operator T on $L^2(\mathbb{R}^n)$ commuting with all $\rho_\lambda(g)$ is a scalar. Write M_φ for multiplication by $\varphi \in L^\infty(\mathbb{R}^n)$.

Step 1: T commutes with every M_φ. The operators $\rho_\lambda(0,b,0) = M_{\mathbf{e}(\lambda t\cdot b)}$ commute with T. If $\phi \in L^1(\mathbb{R}^n)$, then the weak integral $\int \phi(b)\, M_{\mathbf{e}(\lambda t\cdot b)}\, db$ equals $M_{\hat\phi(-\lambda t)}$, so T commutes with $M_{\hat\phi(-\lambda\,\cdot)}$. The functions $\hat\phi(-\lambda\,\cdot)$ with $\phi \in L^1$ are dense in $C_0(\mathbb{R}^n)$ in the uniform norm, and $\|M_\varphi\| = \|\varphi\|_\infty$, so T commutes with M_φ for all $\varphi \in C_0(\mathbb{R}^n)$. An arbitrary $\varphi \in L^\infty$ is an almost everywhere limit of

a uniformly bounded sequence $\varphi_k \in C_c(\mathbb{R}^n)$, and then $M_{\varphi_k} \to M_\varphi$ strongly by dominated convergence; commutation passes to strong limits.

Step 2: T is a multiplication operator. Let $g(t) = e^{-\pi|t|^2}$ and $\theta = Tg/g$, a measurable function. For $\varphi \in L^\infty$ with $\varphi g \in L^2$, Step 1 gives $T(\varphi g) = \varphi\, Tg = \theta\, \varphi g$. Taking φ to be the indicator function of $\{|\theta| > \|T\| + \varepsilon\} \cap \{|t| \le r\}$ divided by g (which is bounded on the ball), the inequality $\|\theta\varphi g\| \le \|T\|\,\|\varphi g\|$ shows this set is null. Hence $\theta \in L^\infty$ and $T = M_\theta$ on the dense subspace $\{\varphi g : \varphi \in L^\infty\}$, so $T = M_\theta$.

Step 3: θ is constant. T also commutes with $\rho_\lambda(a,0,0) = \mathsf{T}_a$, so $\theta(t+a) = \theta(t)$ for almost every t, for each a. For $\phi \in C_c(\mathbb{R}^n)$, Fubini gives $\int \theta(t+a)\phi(a)\,da = \theta(t)\int \phi$ for almost every t. The left-hand side is a continuous function of t that is invariant under all translations, hence constant. Choosing $\int \phi = 1$ shows θ is almost everywhere constant. □

5.3. **The Stone–von Neumann theorem.**

Theorem 5.4 (Stone–von Neumann)**.** *Let $\lambda \neq 0$.*

(1) *Every irreducible unitary representation of H with central character ψ_λ is equivalent to ρ_λ.*
(2) *Every unitary representation of H with central character ψ_λ is equivalent to a Hilbert direct sum of copies of ρ_λ.*

For proofs see [Fol89, Ch. 1], [Tha98, Ch. 1], or the discussion in [How80]. The second statement is the form originally due to von Neumann; it follows from the first together with the fact that every representation with central character ψ_λ contains a copy of ρ_λ, by a Zorn's lemma argument. In this paper Theorem 5.4 is used only once, in Corollary 9.6, to show that no irreducible representations other than the ones we construct by hand occur in $L^2(\Gamma\backslash H)$.

Remark 5.5. Complex conjugation $f \mapsto \bar{f}$ is an antiunitary map intertwining ρ_λ with $\rho_{-\lambda}$. So $\rho_{-\lambda}$ is the complex conjugate of ρ_λ, and nothing below depends essentially on the sign of m.

6. Decomposition by central character

The center Z commutes with H, so its action on $L^2(\Gamma\backslash H)$ can be diagonalized first. Since $(0,0,s)$ is central, right and left multiplication by it agree:

$$(x,y,z)(0,0,s) = (0,0,s)(x,y,z) = (x,y,z+s).$$

Taking $\gamma = (0,0,c) \in \Gamma \cap Z$ in $F(\gamma h) = F(h)$ shows that every $F \in L^2(\Gamma\backslash H)$ is 1-periodic in z.

Definition 6.1. For $m \in \mathbb{Z}$ let

$$\mathcal{H}_m = \big\{F \in L^2(\Gamma\backslash H) : R(0,0,s)F = \mathbf{e}(ms)F \text{ for all } s \in \mathbb{R}\big\}.$$

Equivalently, $F \in \mathcal{H}_m$ if and only if $F(x,y,z) = \mathbf{e}(mz)\varphi(x,y)$ for some measurable φ.

Proposition 6.2. *Each $\mathcal{H}_m$ is a closed H-invariant subspace, and*

$$L^2(\Gamma\backslash H) = \widehat{\bigoplus}_{m\in\mathbb{Z}} \mathcal{H}_m.$$

On $\mathcal{H}_m$ the center acts by the character ψ_m. No character ψ_λ with $\lambda \notin \mathbb{Z}$ occurs.

Proof. $\mathcal{H}_m$ is closed, being an intersection of kernels of the bounded operators $R(0,0,s)-\mathbf{e}(ms)$. It is invariant because $R(g)$ commutes with $R(0,0,s)$.

Define, for $F\in L^2(\Gamma\backslash H)$,

$$(P_mF)(h)=\int_0^1 F\big(h(0,0,s)\big)\,\mathbf{e}(-ms)\,ds.$$

In coordinates, substituting $u=z+s$ and using 1-periodicity in z,

$$(P_mF)(x,y,z)=\int_0^1 F(x,y,z+s)\,\mathbf{e}(-ms)\,ds=\mathbf{e}(mz)\,\varphi_m(x,y), \tag{6.1}$$

where

$$\varphi_m(x,y)=\int_0^1 F(x,y,u)\,\mathbf{e}(-mu)\,du.$$

Thus $\varphi_m(x,y)$ is the m-th Fourier coefficient of $z\mapsto F(x,y,z)$, which by Fubini is defined for almost every (x,y). The function P_mF is left Γ-invariant, because the integrand $F(\gamma h(0,0,s))=F(h(0,0,s))$ is, and by (6.1) it lies in $\mathcal{H}_m$. If $F\in\mathcal{H}_m$ then $P_mF=F$.

For a.e. (x,y), the function $z\mapsto F(x,y,z)$ lies in $L^2(\mathbb{R}/\mathbb{Z})$, and Parseval's identity gives

$$\int_0^1 |F(x,y,z)|^2\,dz=\sum_{m\in\mathbb{Z}}|\varphi_m(x,y)|^2.$$

Integrating over $(x,y)\in[0,1)^{2n}$ and using $\|P_mF\|^2=\int_{[0,1)^{2n}}|\varphi_m|^2$ gives $\|F\|^2=\sum_m\|P_mF\|^2$. The subspaces $\mathcal{H}_m$ are mutually orthogonal: if $F=\mathbf{e}(mz)\varphi\in\mathcal{H}_m$ and $G=\mathbf{e}(m'z)\psi\in\mathcal{H}_{m'}$ with $m\neq m'$, then integrating first in z over D gives

$$\langle F,G\rangle=\int_{[0,1)^{2n}}\varphi\bar{\psi}\,dx\,dy\cdot\int_0^1\mathbf{e}\big((m-m')z\big)\,dz=0.$$

Together these show $F=\sum_m P_mF$ in L^2 and the direct sum decomposition.

Finally, if a character ψ_λ occurs, i.e., $R(0,0,s)F=\mathbf{e}(\lambda s)F$ for some $F\neq0$, then periodicity in z forces $\mathbf{e}(\lambda)=1$, so $\lambda\in\mathbb{Z}$. □

The last statement is the first instance of a general principle: the lattice cuts down the continuous family of representations of H to a discrete set. Concretely, a central character can only occur if it is trivial on $\Gamma\cap Z$.

7. The abelian part

For $(a,b)\in\mathbb{Z}^{2n}$ define $\chi_{a,b}(x,y,z)=\mathbf{e}(a\cdot x+b\cdot y)$.

Proposition 7.1. *The functions $\chi_{a,b}$, $(a,b)\in\mathbb{Z}^{2n}$, form an orthonormal basis of $\mathcal{H}_0$, and each spans an invariant line on which H acts by the character $(p,q,r)\mapsto\mathbf{e}(a\cdot p+b\cdot q)$. Thus*

$$\mathcal{H}_0=\widehat{\bigoplus}_{(a,b)\in\mathbb{Z}^{2n}}\mathbb{C}\chi_{a,b},$$

and these are pairwise inequivalent one-dimensional representations. Moreover, they are exactly the unitary characters of H that are trivial on Γ.

Proof. If $F \in \mathcal{H}_0$ then $F(x,y,z) = \varphi(x,y)$, and by (4.1) left Γ-invariance reads $\varphi(x+a, y+b) = \varphi(x,y)$ for all $a, b \in \mathbb{Z}^n$; the variable z, which carries all the noncommutativity of H, has disappeared. Since $\|F\|^2 = \int_{[0,1)^{2n}} |\varphi|^2$, this identifies $\mathcal{H}_0$ unitarily with $L^2(\mathbb{T}^{2n})$, of which the $\chi_{a,b}$ form an orthonormal basis. For $g = (p,q,r)$,

$$(R(g)\chi_{a,b})(x,y,z) = \chi_{a,b}(x+p,\ y+q,\ z+r+x\cdot q) = \mathbf{e}(a\cdot p + b\cdot q)\,\chi_{a,b}(x,y,z).$$

Distinct (a,b) give distinct characters. Finally, any unitary character of H is trivial on $[H,H] = Z$ (Proposition 3.2), so it factors through $H/Z = \mathbb{R}^{2n}$ and has the form $(x,y,z) \mapsto \mathbf{e}(\alpha\cdot x + \beta\cdot y)$; it is trivial on Γ if and only if $\alpha, \beta \in \mathbb{Z}^n$. □

Geometrically, $\mathcal{H}_0$ consists of the Z-invariant functions, i.e., the functions on the base of the circle bundle $\Gamma\backslash H \to \Gamma\backslash H/Z = \mathbb{T}^{2n}$, and its decomposition is ordinary Fourier analysis on the torus.

8. The components with $m \neq 0$

Fix $m \neq 0$. In this section we determine $\mathcal{H}_m$ as a Hilbert space; the action of H is identified in Section 9.

8.1. **Invariance as twisted periodicity.** Write $F \in \mathcal{H}_m$ as $F(x,y,z) = \mathbf{e}(mz)\varphi(x,y)$. For $\gamma = (a,b,c) \in \Gamma$, (4.1) gives

$$F(\gamma h) = \mathbf{e}\big(m(z+c+a\cdot y)\big)\,\varphi(x+a,y+b) = \mathbf{e}(mz)\,\mathbf{e}(m\,a\cdot y)\,\varphi(x+a,y+b),$$

since $\mathbf{e}(mc) = 1$. Hence F is left Γ-invariant if and only if

$$\varphi(x+a,\ y+b) = \mathbf{e}(-m\,a\cdot y)\,\varphi(x,y) \qquad \text{for all } a,b \in \mathbb{Z}^n, \tag{8.1}$$

and $\|F\|^2 = \int_{[0,1)^{2n}} |\varphi|^2$.

Remark 8.1. The factor $\mathbf{e}(-m\,a\cdot y)$ in (8.1) satisfies the cocycle condition: applying (8.1) for (a,b) and then for (a',b') produces the factor $\mathbf{e}(-m\,a'\cdot(y+b))\,\mathbf{e}(-m\,a\cdot y) = \mathbf{e}(-m(a+a')\cdot y)$, because $\mathbf{e}(-m\,a'\cdot b) = 1$. Geometrically, (8.1) says that φ is an L^2 section of a complex line bundle $\mathcal{L}^{\otimes m}$ on $\mathbb{T}^{2n}$, where $\mathcal{L}$ is the line bundle associated with the circle bundle $\Gamma\backslash H \to \mathbb{T}^{2n}$. See [Mum91, §3] and [Pol03, Part I].

8.2. **Fourier expansion in y.** Taking $a = 0$ in (8.1) shows that $\varphi(x,\cdot)$ is $\mathbb{Z}^n$-periodic, so for almost every x we may expand

$$\varphi(x,y) = \sum_{k\in\mathbb{Z}^n} c_k(x)\,\mathbf{e}(k\cdot y), \qquad c_k(x) = \int_{[0,1)^n} \varphi(x,y)\,\mathbf{e}(-k\cdot y)\,dy. \tag{8.2}$$

Taking $b = 0$ in (8.1) and computing Fourier coefficients of the translate gives

$$c_k(x+a) = \int_{[0,1)^n} \varphi(x,y)\,\mathbf{e}(-m\,a\cdot y)\,\mathbf{e}(-k\cdot y)\,dy = c_{k+ma}(x) \qquad (a \in \mathbb{Z}^n). \tag{8.3}$$

This identity is the heart of the matter: *translating x by $a \in \mathbb{Z}^n$ shifts the Fourier index by ma.* All the noncommutativity of H is concentrated in this shift. It means that the coefficient functions are organized by the residue class of k modulo $m\mathbb{Z}^n$; see Figure 1.

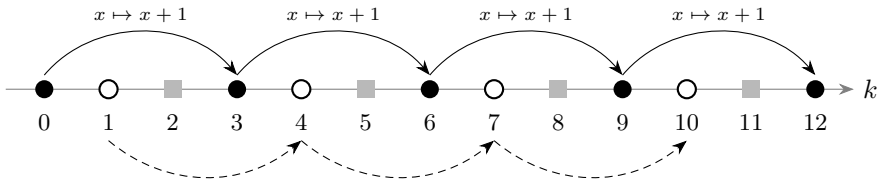


FIGURE 1. The case $n = 1$, $m = 3$. By (8.3), translating x by 1 carries the Fourier coefficient c_k to c_{k+3} (solid arrows for the class of 0, dashed arrows for the class of 1). The indices split into $|m|^n = 3$ residue classes (filled circles, open circles, squares), and each class is described by a single function on $\mathbb{R}$; see (8.4).

8.3. **Reduction to $|m|^n$ functions on $\mathbb{R}^n$.** Let $\Lambda_m = \{0, 1, \dots, |m| - 1\}^n$, a set of representatives for $\mathbb{Z}^n/m\mathbb{Z}^n$, with $|\Lambda_m| = |m|^n$. Every $k \in \mathbb{Z}^n$ can be written uniquely as $k = r + ma$ with $r \in \Lambda_m$ and $a \in \mathbb{Z}^n$. For $r \in \Lambda_m$ set $g_r = c_r$, regarded as a function on all of $\mathbb{R}^n$. Then (8.3) gives $c_{r+ma}(x) = g_r(x + a)$, so

$$\varphi(x, y) = \sum_{r \in \Lambda_m} \sum_{a \in \mathbb{Z}^n} g_r(x + a)\, \mathbf{e}\big((r + ma) \cdot y\big). \tag{8.4}$$

Proposition 8.2. *For $m \neq 0$ the map $U_m \colon F \mapsto (g_r)_{r \in \Lambda_m}$ is a unitary isomorphism*

$$U_m \colon \mathcal{H}_m \xrightarrow{\ \sim\ } \bigoplus_{r \in \Lambda_m} L^2(\mathbb{R}^n).$$

Proof. Isometry. By Parseval in y, Tonelli, and the tiling $\mathbb{R}^n = \bigsqcup_{a \in \mathbb{Z}^n} ([0,1)^n + a)$,

$$\|F\|^2 = \int_{[0,1)^n} \sum_{k \in \mathbb{Z}^n} |c_k(x)|^2 \, dx = \sum_{r \in \Lambda_m} \sum_{a \in \mathbb{Z}^n} \int_{[0,1)^n} |g_r(x+a)|^2 \, dx = \sum_{r \in \Lambda_m} \|g_r\|^2_{L^2(\mathbb{R}^n)}.$$

In particular each g_r lies in $L^2(\mathbb{R}^n)$, and U_m is linear and isometric.

Surjectivity. Let $(g_r)_{r \in \Lambda_m}$ be given, and define φ by (8.4). The function $x \mapsto \sum_{r,a} |g_r(x + a)|^2$ is $\mathbb{Z}^n$-periodic with integral $\sum_r \|g_r\|^2 < \infty$ over $[0,1)^n$, so it is finite for almost every x, and for such x the series (8.4) converges in $L^2(\mathbb{T}^n)$. It is clearly $\mathbb{Z}^n$-periodic in y. For $a' \in \mathbb{Z}^n$, reindexing by $a'' = a + a'$ gives

$$\varphi(x + a', y) = \sum_{r, a''} g_r(x + a'')\, \mathbf{e}\big((r + ma'' - ma') \cdot y\big) = \mathbf{e}(-m\, a' \cdot y)\, \varphi(x, y),$$

so φ satisfies (8.1), and $F = \mathbf{e}(mz)\varphi$ is an element of $\mathcal{H}_m$ with $U_m F = (g_r)$. □

Remark 8.3. The choice of representatives Λ_m is immaterial: replacing r by $r + ma_0$ replaces g_r by its translate $\mathsf{T}_{a_0} g_r$. For $m = 0$, identity (8.3) instead says that each c_k is $\mathbb{Z}^n$-periodic, and we are back in Section 7.

9. The theta map and the main theorem

9.1. **Theta series.** Fix $m \neq 0$. For $f \in \mathcal{S}(\mathbb{R}^n)$ and $j \in \mathbb{Z}^n$ define

$$\Theta_{f,j}(h) = \sum_{k \in \frac{j}{m} + \mathbb{Z}^n} \big(\rho_m(h) f\big)(k) \qquad (h \in H). \tag{9.1}$$

In words: move f by h in the Schrödinger representation, then sum its values over a shifted lattice. Only the class of j modulo $m\mathbb{Z}^n$ matters. For $h = (x, y, z)$ and $k = \frac{j}{m} + a$ we have $m(z + k \cdot y) = mz + (j + ma) \cdot y$, so

$$\Theta_{f,j}(x, y, z) = \mathbf{e}(mz) \sum_{a \in \mathbb{Z}^n} f\Big(x + a + \frac{j}{m}\Big)\, \mathbf{e}\big((j + ma) \cdot y\big). \tag{9.2}$$

Because f is a Schwartz function, the series converges absolutely and uniformly on compact sets together with all its derivatives, so $\Theta_{f,j}$ is a smooth function on H.

Proposition 9.1. *Let $f \in \mathcal{S}(\mathbb{R}^n)$ and $j \in \mathbb{Z}^n$.*

(1) (Automorphy) $\Theta_{f,j}(\gamma h) = \Theta_{f,j}(h)$ *for all* $\gamma \in \Gamma$.
(2) (Equivariance) $R(g)\Theta_{f,j} = \Theta_{\rho_m(g)f,\, j}$ *for all* $g \in H$.
(3) (Central character) $\Theta_{f,j} \in \mathcal{H}_m$.
(4) (Fourier coefficients) *If $r \in \Lambda_m$, then $U_m \Theta_{f,r}$ is the vector whose r-th component is $\mathsf{T}_{r/m} f$ and whose other components are 0. In particular $\|\Theta_{f,r}\| = \|f\|$, and $\Theta_{f,r} \perp \Theta_{f',r'}$ for $r \neq r'$.*

Proof. (1) Let $\gamma = (a, b, c)$ and $k = \frac{j}{m} + a'$. By the homomorphism property,

$$\big(\rho_m(\gamma h) f\big)(k) = \big(\rho_m(\gamma)[\rho_m(h) f]\big)(k) = \mathbf{e}\big(m(c + k \cdot b)\big)\, [\rho_m(h) f](k + a).$$

Here $mc \in \mathbb{Z}$ and $m\, k \cdot b = (j + ma') \cdot b \in \mathbb{Z}$, so the phase equals 1; and $k \mapsto k + a$ permutes the coset $\frac{j}{m} + \mathbb{Z}^n$. Absolute convergence permits the reordering, so the sums agree.

This computation explains the shifted lattices: the cosets of $\mathbb{Z}^n$ in $\frac{1}{m}\mathbb{Z}^n$ are exactly the $\mathbb{Z}^n$-stable sets of evaluation points on which the phase $\mathbf{e}(m\, k \cdot b)$ is trivial for all $b \in \mathbb{Z}^n$.

(2) Since $\rho_m(g)$ preserves $\mathcal{S}(\mathbb{R}^n)$,

$$\Theta_{f,j}(hg) = \sum_k \big(\rho_m(h)\rho_m(g) f\big)(k) = \Theta_{\rho_m(g)f,\, j}(h).$$

(3) By (9.2), $\Theta_{f,j}(x, y, z + s) = \mathbf{e}(ms)\Theta_{f,j}(x, y, z)$, and $\Theta_{f,j}$ is left Γ-invariant by (1); it is bounded on the compact set $\overline{D}$, hence square integrable.

(4) For $j = r \in \Lambda_m$, formula (9.2) is already a Fourier series in y, with coefficients $c_{r+ma}(x) = f(x + a + \frac{r}{m})$ and $c_{r'+ma} = 0$ for $r' \neq r$. Comparing with the definition of U_m gives $g_r = \mathsf{T}_{r/m} f$ and $g_{r'} = 0$. The remaining claims follow because U_m and $\mathsf{T}_{r/m}$ are unitary. □

9.2. **The main theorem.** By Proposition 9.1(4), $f \mapsto \Theta_{f,r}$ is an isometry from the dense subspace $\mathcal{S}(\mathbb{R}^n)$ of $L^2(\mathbb{R}^n)$ into $\mathcal{H}_m$, so it extends uniquely to an isometry

$$\Theta_r \colon L^2(\mathbb{R}^n) \to \mathcal{H}_m.$$

The identity $R(g)\Theta_r = \Theta_r \rho_m(g)$ holds on $\mathcal{S}(\mathbb{R}^n)$ by Proposition 9.1(2); both sides are bounded, so it holds on $L^2(\mathbb{R}^n)$. Thus each Θ_r is an isometric intertwiner from ρ_m into $\mathcal{H}_m$.

Theorem 9.2. *For $m \neq 0$, the map*

$$\Theta = \sum_{r \in \Lambda_m} \Theta_r \colon \bigoplus_{r \in \Lambda_m} \big(\rho_m, L^2(\mathbb{R}^n)\big) \longrightarrow \mathcal{H}_m$$

is a unitary isomorphism of representations of H. In particular $\mathcal{H}_m \cong \rho_m^{\oplus |m|^n}$.

Proof. Θ is an intertwiner, and it is isometric because the images of the Θ_r are mutually orthogonal. It remains to show it is surjective. Let $F \in \mathcal{H}_m$ with $U_m F = (g_r)$, and put $f_r = \mathsf{T}_{-r/m} g_r$. By Proposition 9.1(4) and continuity, $U_m\big(\sum_r \Theta_r f_r\big) = (g_r) = U_m F$, and U_m is injective, so $F = \sum_r \Theta_r f_r$. □

Combining Theorem 9.2 with Propositions 6.2 and 7.1 gives the Main Theorem stated in the introduction:

Theorem 9.3. *Under the right regular representation of $H = H_n(\mathbb{R})$,*

$$L^2(H_n(\mathbb{Z})\backslash H_n(\mathbb{R})) \;\cong\; \widehat{\bigoplus}_{(a,b)\in\mathbb{Z}^{2n}} \chi_{a,b} \;\oplus\; \widehat{\bigoplus}_{m\in\mathbb{Z}\setminus\{0\}} \rho_m^{\oplus |m|^n},$$

and on the m-th piece the isomorphism is the theta map Θ of Theorem 9.2.

9.3. **The Weil–Brezin map.** For $m = 1$ there is a single residue class, and $\Theta = \Theta_0$ is the map

(9.3)

$$(Wf)(x,y,z) = \mathbf{e}(z) \sum_{k\in\mathbb{Z}^n} f(x+k)\,\mathbf{e}(k\cdot y), \qquad (W^{-1}F)(x) = \int_{[0,1)^n} F(x,y,0)\,dy.$$

The inversion formula expresses f as the 0-th Fourier coefficient in y: integrating (9.3) term by term over y kills every $k \neq 0$. By Theorem 9.2, W is a unitary H-isomorphism from $(\rho_1, L^2(\mathbb{R}^n))$ onto $\mathcal{H}_1$. This is the *Weil–Brezin map* [Bre70, AT75, Aus77]. Outside representation theory, the function $(x,y) \mapsto \sum_k f(x+k)\mathbf{e}(k\cdot y)$ is known as the *Zak transform* of f, and it is a basic tool in time-frequency analysis [Grö01].

9.4. **Irreducibility, multiplicities, and completeness.** Theorem 9.3 was proved by explicit construction. Combined with the irreducibility of ρ_m (Proposition 5.3) and Schur's lemma, it yields three further statements. Only the last of them uses the Stone–von Neumann theorem.

Corollary 9.4. *Each subspace $\Theta_r(L^2(\mathbb{R}^n))$ is irreducible. In particular $\mathcal{H}_1$ and $\mathcal{H}_{-1}$ are irreducible.*

Proof. It is equivalent to ρ_m, which is irreducible by Proposition 5.3. □

Corollary 9.5. *For $m \neq 0$, the intertwiners Θ_r with $r \in \Lambda_m$ form a basis of the space* $\mathrm{Hom}_H\big(\rho_m, L^2(\Gamma\backslash H)\big)$. *In particular*

$$\dim \mathrm{Hom}_H\big(\rho_m, L^2(\Gamma\backslash H)\big) = |m|^n.$$

Proof. If T is an intertwiner from ρ_m to $L^2(\Gamma\backslash H)$, then for all f and s we have $R(0,0,s)Tf = T\rho_m(0,0,s)f = \mathbf{e}(ms)Tf$, so T takes values in $\mathcal{H}_m$. Now apply Lemma 2.5 to the decomposition of Theorem 9.2. □

Corollary 9.6. *Every irreducible closed invariant subspace of $L^2(\Gamma\backslash H)$ is either one of the lines $\mathbb{C}\chi_{a,b}$ or is equivalent to ρ_m for some $m \neq 0$.*

Proof. Let V be such a subspace. By Corollary 2.4, Z acts on V by a single character, which is ψ_m for some $m \in \mathbb{Z}$ by Proposition 6.2, so $V \subseteq \mathcal{H}_m$. If $m = 0$, then H acts on V through the abelian group H/Z, so every operator $R(g)|_V$ commutes with all of $R(H)|_V$ and is a scalar by Lemma 2.3; hence every closed subspace of V is invariant and V is one-dimensional. A spanning vector transforms by a character of H trivial on Γ, so by Proposition 7.1 it is a multiple of some $\chi_{a,b}$. If $m \neq 0$, apply Theorem 5.4(1). □

9.5. **The multiplicity space and a finite Heisenberg group.** The isotypic components $\mathcal{H}_m$ are canonical, but for $|m|^n > 1$ the splitting of $\mathcal{H}_m$ into irreducibles is not. By Corollaries 9.5 and 9.6, the irreducible subspaces of $\mathcal{H}_m$ are exactly the images of the intertwiners $\sum_r \lambda_r \Theta_r$ with $(\lambda_r) \neq 0$. The invariant statement is that evaluation $T \otimes f \mapsto Tf$ is an isomorphism of H-representations

$$\mathrm{Hom}_H(\rho_m, \mathcal{H}_m) \otimes \rho_m \xrightarrow{\sim} \mathcal{H}_m,$$

where H acts on the second factor only. We now show that the $|m|^n$-dimensional *multiplicity space* $\mathrm{Hom}_H(\rho_m, \mathcal{H}_m)$ carries a natural irreducible action of a finite Heisenberg group. This gives a second explanation of the number $|m|^n$, and it shows that no copy of ρ_m inside $\mathcal{H}_m$ is distinguished.

Fix $m \neq 0$. For $u, v \in \mathbb{Z}^n$ put $\nu_{u,v} = (u/m,\, v/m,\, 0) \in H$, and define

$$(L_{u,v}F)(h) = F(\nu_{u,v}^{-1} h).$$

Proposition 9.7. *Let $m \neq 0$ and $u, v \in \mathbb{Z}^n$.*

(1) *$L_{u,v}$ is a unitary operator on $\mathcal{H}_m$ that commutes with $R(g)$ for every $g \in H$.*
(2) *For $f \in \mathcal{S}(\mathbb{R}^n)$ and $j \in \mathbb{Z}^n$,*

$$L_{u,0}\Theta_{f,j} = \Theta_{f,j-u}, \qquad L_{0,v}\Theta_{f,j} = \mathbf{e}(-j \cdot v/m)\, \Theta_{f,j}.$$

(3) *The only subspaces of $\mathrm{Hom}_H(\rho_m, \mathcal{H}_m)$ that are invariant under all the maps $T \mapsto L_{u,v} \circ T$ are 0 and the whole space.*

Proof. (1) Write $\nu = \nu_{u,v}$, so $\nu^{-1} = (-u/m, -v/m, u \cdot v/m^2)$, and let $\gamma = (a, b, c) \in \Gamma$. By (3.3), $\nu^{-1}\gamma = \gamma\nu^{-1} \cdot (0, 0, \omega)$ with $\omega = (a \cdot v - u \cdot b)/m$. Since $(0, 0, \omega)$ is central and $F \in \mathcal{H}_m$,

$$F(\nu^{-1}\gamma h) = F\big(\gamma\nu^{-1} h\,(0, 0, \omega)\big) = \mathbf{e}(m\omega)\, F(\gamma\nu^{-1}h) = F(\nu^{-1}h),$$

because $m\omega \in \mathbb{Z}$ and F is left Γ-invariant. So $L_{u,v}F$ is left Γ-invariant. It lies in $\mathcal{H}_m$, and $L_{u,v}$ commutes with every $R(g)$, because left and right translations commute. For unitarity, write $F = \mathbf{e}(mz)\varphi(x, y)$. By (8.1), $|F|^2 = |\varphi|^2$ is $\mathbb{Z}^{2n}$-periodic in (x, y), and $L_{u,v}$ translates (x, y) by $(-u/m, -v/m)$, so $\|L_{u,v}F\| = \|F\|$. The inverse of $L_{u,v}$ is left translation by ν, which has the same properties.

(2) By the homomorphism property, $\Theta_{f,j}(\nu^{-1}h) = \sum_k \big(\rho_m(\nu^{-1})[\rho_m(h)f]\big)(k)$, the sum running over $k \in \frac{j}{m} + \mathbb{Z}^n$. For $\nu = \nu_{u,0}$ we have $\nu^{-1} = (-u/m, 0, 0)$ and $(\rho_m(\nu^{-1})\phi)(k) = \phi(k - u/m)$; the map $k \mapsto k - u/m$ carries $\frac{j}{m} + \mathbb{Z}^n$ onto $\frac{j-u}{m} + \mathbb{Z}^n$. For $\nu = \nu_{0,v}$ we have $\nu^{-1} = (0, -v/m, 0)$ and $(\rho_m(\nu^{-1})\phi)(k) = \mathbf{e}(-k \cdot v)\phi(k)$, and $\mathbf{e}(-k \cdot v) = \mathbf{e}(-j \cdot v/m)$ for every $k \in \frac{j}{m} + \mathbb{Z}^n$.

(3) Since $\Theta_{f,j}$ depends only on j modulo $m\mathbb{Z}^n$, we may index the basis of Corollary 9.5 by $j \in \mathbb{Z}^n/m\mathbb{Z}^n$. Both sides of the identities in (2) are bounded in f, so they hold with Θ_j in place of $\Theta_{f,j}$. Identifying $\mathrm{Hom}_H(\rho_m, \mathcal{H}_m)$ with $\mathbb{C}^{\mathbb{Z}^n/m\mathbb{Z}^n}$ via $\lambda \mapsto \sum_j \lambda_j \Theta_j$, the map $T \mapsto L_{0,v} \circ T$ becomes multiplication by the function $j \mapsto \mathbf{e}(-j \cdot v/m)$, and $T \mapsto L_{u,0} \circ T$ becomes the shift $(\lambda_j) \mapsto (\lambda_{j+u})$. Let $W \neq 0$ be an invariant subspace. The characters $j \mapsto \mathbf{e}(-j \cdot v/m)$ of the finite abelian group $\mathbb{Z}^n/m\mathbb{Z}^n$ span all functions on it, so W is stable under multiplication by every function, in particular by each indicator δ_{j_0}. Choosing $\lambda \in W$ with $\lambda_{j_0} \neq 0$ shows $\delta_{j_0} \in W$, and the shifts then give every δ_j. So W is the whole space. □

Remark 9.8. The operators $L_{u,v}$ depend only on (u, v) modulo $m\mathbb{Z}^{2n}$, since $\nu_{u,v} \in \Gamma$ when $u, v \in m\mathbb{Z}^n$. By (2), $L_{u,0}L_{0,v} = \mathbf{e}(-u \cdot v/m)\, L_{0,v}L_{u,0}$, and $L_{u,v}$ differs from $L_{u,0}L_{0,v}$ by a root of unity. So, together with the group $\mu_{|m|}$ of $|m|$-th roots of unity,

the $L_{u,v}$ generate a finite Heisenberg group: a central extension of $(\mathbb{Z}/m\mathbb{Z})^{2n}$ by $\mu_{|m|}$ whose commutator pairing $\big((u,v),(u',v')\big) \mapsto \mathbf{e}\big((u'\cdot v - u\cdot v')/m\big)$ is nondegenerate. Proposition 9.7(3) is the (elementary) finite analogue of the Stone–von Neumann theorem: up to equivalence, this group has a unique irreducible representation on which $\mu_{|m|}$ acts by scalar multiplication, and it has dimension $|m|^n$ [Mum91, §1]. In this sense the multiplicity of ρ_m in $L^2(\Gamma\backslash H)$ is the dimension of a finite Schrödinger representation, and

$$\mathcal{H}_m \cong \mathbb{C}^{\mathbb{Z}^n/m\mathbb{Z}^n} \otimes \rho_m$$

as a representation of the product of the finite Heisenberg group (acting on the first factor) and H (acting on the second). Since the first factor is irreducible, no nonzero proper subspace of the multiplicity space, and hence no particular copy of ρ_m, is singled out by the symmetries of $\Gamma\backslash H$. This finite Heisenberg group is the starting point of the finite Fourier transform in [Aus77, Ch. II] and of Mumford's algebraic theory of theta functions [Mum66, Mum91]; Section 11 gives it an adelic interpretation.

Remark 9.9 (Multiplicity and formal degree). The representation ρ_m is square integrable modulo the center. Indeed, for $u, v \in L^2(\mathbb{R}^n)$ we have $\langle \rho_m(x,y,0)u, v\rangle = \int \mathbf{e}(m\,t\cdot y)\, u(t+x)\overline{v(t)}\, dt$, so Plancherel's theorem in y, followed by Fubini in (t,x), gives

$$\int_{\mathbb{R}^{2n}} \big|\langle \rho_m(x,y,0)u, v\rangle\big|^2 \, dx\, dy = \frac{\|u\|^2\,\|v\|^2}{|m|^n}.$$

Thus the formal degree of ρ_m, computed with the Haar measure $dx\,dy$ on $H/Z = \mathbb{R}^{2n}$, is $|m|^n$. This measure gives the image $\mathbb{Z}^{2n}$ of Γ covolume 1. Moore and Wolf proved that for a nilpotent Lie group with a lattice, normalized in this way, a representation that is square integrable modulo the center occurs in $L^2(\Gamma\backslash G)$ if and only if its central character is trivial on $\Gamma\cap Z$, and then its multiplicity equals its formal degree [MW73, §6, Thm. 7]. For the Heisenberg group this recovers Corollary 9.5. It is a noncompact analogue of the fact, from the Peter–Weyl theorem, that each irreducible representation of a compact group occurs in its regular representation with multiplicity equal to its dimension. Other choices of lattice change the count accordingly. For instance, [Tha09, §3.2] works in the symmetric model (Remark 3.3) with the lattice $\mathbb{Z}^{2n}\times\frac{1}{2}\mathbb{Z}$, whose central characters are $t\mapsto e^{4\pi i k t}$ with $k\in\mathbb{Z}$, i.e., ψ_{2k} in our notation; the formal degree is then $|2k|^n$, and accordingly [Tha09, Remark 3.1] finds multiplicity $(2|k|)^n$.

Remark 9.10 (Invariant functionals). Evaluating (9.1) at the identity gives the tempered distribution $\theta_j(f) = \sum_{k\in j/m+\mathbb{Z}^n} f(k)$, and the definition reads

$$\Theta_{f,j}(h) = \theta_j\big(\rho_m(h)f\big), \qquad \theta_j\big(\rho_m(\gamma)f\big) = \theta_j(f) \quad (\gamma\in\Gamma).$$

This is the general recipe for automorphic forms: intertwiners from an irreducible representation π into $L^2(\Gamma\backslash G)$ correspond to Γ-invariant continuous functionals on the smooth vectors of π. From this point of view Corollary 9.5 says that the Γ-invariant tempered distributions for ρ_m are spanned by the $|m|^n$ coset sums θ_j. This viewpoint goes back to Cartier [Car66]; see also [LV80, Ch. 1].

10. Theta functions

We now explain why the vectors constructed above deserve to be called automorphic forms: they include the classical theta functions, and the representation theory of Section 9 proves their basic properties.

10.1. **Gaussians and the Riemann theta function.** Let $\mathfrak{S}_n$ be the Siegel upper half space of complex symmetric $n \times n$ matrices Ω with positive definite imaginary part; for $n = 1$ it is the upper half plane $\mathfrak{H}$. For $\Omega \in \mathfrak{S}_n$, the Gaussian $f_\Omega(t) = e^{\pi i\, t\cdot \Omega t}$ is a Schwartz function, and the *Riemann theta function* is

$$\theta(w,\Omega) = \sum_{k\in\mathbb{Z}^n} e^{\pi i\, k\cdot\Omega k + 2\pi i\, k\cdot w} \qquad (w \in \mathbb{C}^n).$$

For $n = 1$ we write $\Omega = \tau$ and $\vartheta(w,\tau) = \theta(w,\tau)$, the Jacobi theta function.

Proposition 10.1. *For $\Omega \in \mathfrak{S}_n$,*

$$(W f_\Omega)(x,y,z) = \mathbf{e}(z)\, e^{\pi i\, x\cdot\Omega x}\, \theta(\Omega x + y, \Omega).$$

Proof. Since Ω is symmetric, $\pi i(x+k)\cdot\Omega(x+k) = \pi i\, x\cdot\Omega x + \pi i\, k\cdot\Omega k + 2\pi i\, k\cdot\Omega x$. Substituting into (9.3),

$$(W f_\Omega)(x,y,z) = \mathbf{e}(z)\, e^{\pi i\, x\cdot\Omega x} \sum_{k\in\mathbb{Z}^n} e^{\pi i\, k\cdot\Omega k + 2\pi i\, k\cdot(\Omega x + y)}. \qquad \square$$

10.2. **Quasi-periodicity is the group law.**

Proposition 10.2. *For $w \in \mathbb{C}^n$ and $a, b \in \mathbb{Z}^n$,*

$$\theta(w+b,\Omega) = \theta(w,\Omega), \qquad \theta(w+\Omega a,\Omega) = e^{-\pi i\, a\cdot\Omega a - 2\pi i\, a\cdot w}\, \theta(w,\Omega).$$

Proof. By Proposition 10.1, $\varphi(x,y) = e^{\pi i\, x\cdot\Omega x}\theta(\Omega x + y, \Omega)$ satisfies the twisted periodicity (8.1) with $m = 1$, because $W f_\Omega \in \mathcal{H}_1$. Put $w = \Omega x + y$; as (x,y) ranges over $\mathbb{R}^{2n}$, w ranges over $\mathbb{C}^n$, since $\operatorname{Im}\Omega$ is invertible. Taking $a = 0$ in (8.1) gives the first identity. Taking $b = 0$ gives

$$e^{\pi i(x+a)\cdot\Omega(x+a)}\, \theta(w+\Omega a,\Omega) = \mathbf{e}(-a\cdot y)\, e^{\pi i\, x\cdot\Omega x}\, \theta(w,\Omega),$$

and solving, using $-2\pi i\, a\cdot\Omega x - 2\pi i\, a\cdot y = -2\pi i\, a\cdot w$, gives the second. $\square$

Of course both identities can also be checked by reindexing the series. The point of the proof above is conceptual: the automorphy factor $e^{-\pi i\, a\cdot\Omega a - 2\pi i\, a\cdot w}$, which looks mysterious in isolation, is the factor $\mathbf{e}(-m\, a\cdot y)$ of (8.1), i.e., the cocycle defining the Heisenberg group.

10.3. **Poisson summation and the transformation law from multiplicity one.** Let J be the automorphism (3.5), and define $J^*F = F\circ J$.

Lemma 10.3. *$\mathfrak{F}\rho_1(h) = \rho_1(J(h))\mathfrak{F}$ for all $h \in H$.*

Proof. For $f \in \mathcal{S}(\mathbb{R}^n)$ and $h = (x,y,z)$, substituting $u = t + x$,

$$\mathfrak{F}\big(\rho_1(h)f\big)(\xi) = \int \mathbf{e}(z + t\cdot y) f(t+x)\mathbf{e}(-t\cdot\xi)\, dt = \mathbf{e}(z - x\cdot y + \xi\cdot x)\, \hat{f}(\xi - y).$$

Since $J(h) = (-y, x, z - x\cdot y)$, the right-hand side is $\big(\rho_1(J(h))\hat{f}\big)(\xi)$ by (5.1). Both sides are bounded operators, so the identity extends to L^2. $\square$

Theorem 10.4. *For every $g \in \mathcal{S}(\mathbb{R}^n)$,*

$$\sum_{k\in\mathbb{Z}^n} g(k) = \sum_{k\in\mathbb{Z}^n} \hat{g}(k).$$

Consequently, for $\Omega \in \mathfrak{S}_n$,

$$\theta(0, -\Omega^{-1}) = \det(-i\Omega)^{1/2}\, \theta(0, \Omega),$$

where the square root is the branch that is positive when Ω is purely imaginary. For $n = 1$ this is $\vartheta(0, -1/\tau) = \sqrt{-i\tau}\, \vartheta(0, \tau)$.

Proof. By Lemma 3.5, J preserves Γ and Haar measure and fixes Z pointwise, so J^* is a unitary operator on $L^2(\Gamma\backslash H)$ preserving $\mathcal{H}_1$ (for unitarity, $\int_D |F \circ J|^2 = \int_{J(D)} |F|^2$, and $J(D)$ is again a fundamental domain because $J(\Gamma) = \Gamma$, so Lemma 4.3 applies). Moreover $R(g)J^* = J^*R(J(g))$, since $F(J(hg)) = F(J(h)J(g))$. Using Lemma 10.3,

$$R(g)\, J^*W\mathfrak{F} = J^*R(J(g))W\mathfrak{F} = J^*W\rho_1(J(g))\mathfrak{F} = J^*W\mathfrak{F}\rho_1(g).$$

Thus $J^*W\mathfrak{F}$ is an intertwiner from ρ_1 to $\mathcal{H}_1$. By Corollary 9.5, with $|m|^n = 1$, it is a scalar multiple cW of W, and $|c| = 1$ because both are isometries. Equivalently,

$$(Wg) \circ J = c\, W(\mathfrak{F}^{-1}g) \qquad (g \in L^2(\mathbb{R}^n)).$$

For $g \in \mathcal{S}(\mathbb{R}^n)$, both sides are continuous functions on H, so the identity holds pointwise. Evaluating at the identity element $1 = (0, 0, 0)$, and using $J(1) = 1$ and $(Wg)(1) = \sum_k g(k)$, gives

$$\sum_k g(k) = c \sum_k (\mathfrak{F}^{-1}g)(k) = c \sum_k \hat{g}(-k) = c \sum_k \hat{g}(k).$$

Testing with $g(t) = e^{-\pi|t|^2}$, which equals its own Fourier transform and has $\sum_k g(k) > 0$, gives $c = 1$.

For the second statement, apply Poisson summation to $g = f_\Omega$. The standard Gaussian integral gives $\hat{f}_\Omega(\xi) = \det(-i\Omega)^{-1/2} e^{-\pi i\, \xi\cdot\Omega^{-1}\xi}$, with the stated branch [Mum83, Ch. I], and hence $\theta(0, \Omega) = \det(-i\Omega)^{-1/2}\theta(0, -\Omega^{-1})$. □

Remark 10.5. Evaluating the identity $(Wg) \circ J = W(\mathfrak{F}^{-1}g)$ at a general point h, rather than at 1, yields the full transformation law $\vartheta(w/\tau, -1/\tau) = \sqrt{-i\tau}\, e^{\pi i w^2/\tau}\vartheta(w, \tau)$. More generally, the symplectic group acts on H by automorphisms, and by the Stone–von Neumann theorem each automorphism σ is implemented on $L^2(\mathbb{R}^n)$ by a unitary operator $\omega(\sigma)$, unique up to a scalar. This projective representation ω is the *Weil* (or metaplectic, or oscillator) representation [Wei64]; see [Fol89, Ch. 4] and [LV80]. The argument above, applied to those σ that preserve Γ, produces the transformation laws of $\theta(w, \Omega)$ under a subgroup of $Sp_{2n}(\mathbb{Z})$ (a "theta group"); the remaining elements of $Sp_{2n}(\mathbb{Z})$ move Γ to lattices that lead to theta functions with characteristics.

10.4. **Theta functions of level m.** Finally we explain the count $|m|^n$ from the classical side. We take $n = 1$ to keep notation light and comment on general n at the end. Fix τ with $\operatorname{Im}\tau > 0$ and $m \geq 1$.

Definition 10.6. The space of *theta functions of level m* is

$$\mathrm{Th}_m(\tau) = \big\{G\colon \mathbb{C} \to \mathbb{C} \text{ entire}:\ G(w{+}1) = G(w),\ G(w{+}\tau) = e^{-\pi i m\tau - 2\pi i m w} G(w)\big\}.$$

For $m = 1$, Proposition 10.2 shows $\vartheta(\cdot,\tau) \in \mathrm{Th}_1(\tau)$. On $\Gamma\backslash H$ consider the operator built from the vector fields (3.4),

$$D_\tau = X - \tau Y = \partial_x - \tau(\partial_y + x\,\partial_z).$$

Lemma 10.7. *The map*

$$G \longmapsto F_G, \qquad F_G(x,y,z) = \mathbf{e}(mz)\, e^{\pi i m\tau x^2}\, G(\tau x + y),$$

is a linear bijection from $\mathrm{Th}_m(\tau)$ *onto the space of smooth functions* $F \in \mathcal{H}_m$ *with* $D_\tau F = 0$.

Proof. Write a smooth $F \in \mathcal{H}_m$ as $F = \mathbf{e}(mz)e^{\pi i m\tau x^2}G(x,y)$ with G smooth on $\mathbb{R}^2$. Then

$$\begin{aligned} D_\tau F &= \mathbf{e}(mz)e^{\pi i m\tau x^2}\big(2\pi i m\tau x\, G + \partial_x G - \tau\partial_y G - 2\pi i m\tau x\, G\big) \\ &= \mathbf{e}(mz)e^{\pi i m\tau x^2}(\partial_x - \tau\partial_y)G. \end{aligned}$$

In the coordinate $w = \tau x + y$, which identifies $\mathbb{R}^2$ with $\mathbb{C}$ because $\operatorname{Im}\tau \neq 0$, we have $\partial_x - \tau\partial_y = (\bar\tau - \tau)\partial_{\bar w}$. So $D_\tau F = 0$ if and only if G is a holomorphic function of w. For such G, condition (8.1) with $(a,b) = (0,1)$ says $G(w+1) = G(w)$, and with $(a,b) = (1,0)$ it says

$$e^{\pi i m\tau(x+1)^2}G(w+\tau) = e^{-2\pi i m y}e^{\pi i m\tau x^2}G(w), \quad \text{i.e. } G(w+\tau) = e^{-\pi i m\tau - 2\pi i m w}G(w).$$

Since $(0,1)$ and $(1,0)$ together with $Z \cap \Gamma$ generate Γ, these two conditions are equivalent to Γ-invariance. □

Theorem 10.8. *For* $m \geq 1$, $\dim \mathrm{Th}_m(\tau) = m$. *A basis is given by the theta functions with characteristics*

$$G_r(w) = \sum_{\ell \in \frac{r}{m}+\mathbb{Z}} e^{\pi i m\tau\ell^2 + 2\pi i m\ell w} \qquad (r = 0, 1, \dots, m-1),$$

and $F_{G_r} = \Theta_r(f)$ *for the Gaussian* $f(t) = e^{\pi i m\tau t^2}$.

Proof. Upper bound. If $G \in \mathrm{Th}_m(\tau)$, periodicity gives $G(w) = \sum_{k\in\mathbb{Z}} a_k \mathbf{e}(kw)$. Substituting into the second functional equation and comparing coefficients,

$$a_k\mathbf{e}(k\tau) = a_{k+m}e^{-\pi i m\tau}, \qquad \text{i.e.,} \qquad a_{k+m} = e^{\pi i(2k+m)\tau}a_k.$$

So G is determined by $a_0, \dots, a_{m-1}$, and $\dim \mathrm{Th}_m(\tau) \leq m$.

Lower bound, via representation theory. Let $f(t) = e^{\pi i m\tau t^2}$; it is Schwartz since $m \operatorname{Im}\tau > 0$. By (5.2), $d\rho_m(X - \tau Y)f = f' - 2\pi i m\tau t f = 0$. Each Θ_r is a bounded intertwiner, so differentiating $R(\exp sV)\Theta_r f = \Theta_r \rho_m(\exp sV)f$ at $s = 0$ gives $D_\tau \Theta_r f = \Theta_r\, d\rho_m(X - \tau Y)f = 0$. (The left side may be differentiated pointwise, since the series (9.2) converges with all derivatives; the right side in L^2; the two derivatives agree almost everywhere.) The m functions $\Theta_r f$, $r \in \Lambda_m$, are linearly independent by Proposition 9.1(4), smooth, lie in $\mathcal{H}_m$, and are killed by D_τ. By Lemma 10.7 they come from m linearly independent elements of $\mathrm{Th}_m(\tau)$.

Identification. From (9.2) with $\ell = a + \frac{r}{m}$,

$$\begin{aligned} \Theta_r f(x,y,z) &= \mathbf{e}(mz) \sum_{\ell \in \frac{r}{m}+\mathbb{Z}} e^{\pi i m\tau(x+\ell)^2}\, \mathbf{e}(m\ell y) \\ &= \mathbf{e}(mz)\, e^{\pi i m\tau x^2} \sum_{\ell \in \frac{r}{m}+\mathbb{Z}} e^{\pi i m\tau\ell^2 + 2\pi i m\ell(\tau x + y)}, \end{aligned}$$

which is F_{G_r}. □

Remark 10.9 (Riemann–Roch). Let $E_\tau = \mathbb{C}/(\mathbb{Z}+\mathbb{Z}\tau)$. The functional equations defining $\mathrm{Th}_m(\tau)$ say precisely that $\mathrm{Th}_m(\tau)$ is the space of holomorphic sections of $\mathcal{L}^{\otimes m}$, where $\mathcal{L}$ is the line bundle of degree one whose sections are the level-one theta functions. Riemann–Roch on the elliptic curve E_τ gives $h^0(E_\tau, \mathcal{L}^{\otimes m}) = \deg \mathcal{L}^{\otimes m} = m$. Theorem 10.8 recovers this from representation theory: there are m copies of ρ_m in $L^2(\Gamma\backslash H)$, and each contains a one-dimensional space of solutions of $D_\tau F = 0$, namely the Gaussians. The same argument in dimension n, with D_Ω given by the n operators $X_j - \sum_k \Omega_{jk} Y_k$, shows that the space of level-m theta functions for $\Omega \in \mathfrak{S}_n$ has dimension m^n, the classical count for a principally polarized abelian variety [Mum91, §3], [Igu72]. Operators of this type, and the interpretation of their solutions as theta functions, are the subject of [Tol78].

11. The adelic point of view

The quotient $\Gamma\backslash H = H_n(\mathbb{Z})\backslash H_n(\mathbb{R})$ is a real manifold, not an adelic object. But it is the shadow of an adelic quotient $H(\mathbb{Q})\backslash H(\mathbb{A})$, exactly as $SL_2(\mathbb{Z})\backslash SL_2(\mathbb{R})$ is the shadow of $SL_2(\mathbb{Q})\backslash SL_2(\mathbb{A})$. In this section we make the relation precise and use it to explain the multiplicity $|m|^n$: adelically every Schrödinger representation occurs with multiplicity one, and the factor $|m|^n$ is contributed by the finite places. Standard references for adeles are [Tat67, Wei67, RV99]; the adelic treatment of the Heisenberg group goes back to Weil [Wei64], and is developed in [Mum91, Ch. I] and, for general nilmanifolds, in [Fox89].

11.1. **Adeles.** For a prime p, let $\mathbb{Q}_p$ be the p-adic numbers and $\mathbb{Z}_p$ the p-adic integers. The ring of *finite adeles* $\mathbb{A}_f$ is the restricted product of the $\mathbb{Q}_p$ with respect to the $\mathbb{Z}_p$: the set of sequences $(x_p)_p$ with $x_p \in \mathbb{Q}_p$ and $x_p \in \mathbb{Z}_p$ for all but finitely many p. It is a locally compact ring in which $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$ is a compact open subring. The *adeles* are $\mathbb{A} = \mathbb{R} \times \mathbb{A}_f$. The field $\mathbb{Q}$ embeds diagonally in $\mathbb{A}$ and in $\mathbb{A}_f$.

Lemma 11.1. (1) $\mathbb{A}_f = \mathbb{Q} + \widehat{\mathbb{Z}}$ *and* $\mathbb{Q} \cap \widehat{\mathbb{Z}} = \mathbb{Z}$.
(2) $\mathbb{Q}$ *is a discrete subgroup of* $\mathbb{A}$, *and* $\mathbb{A} = \mathbb{Q} + \big(\mathbb{R} \times \widehat{\mathbb{Z}}\big)$.

Proof. (1) For $x_p \in \mathbb{Q}_p$ let $\{x_p\}_p$ denote its *principal part*: if $x_p = \sum_{i \geq -k} c_i p^i$ with digits $0 \leq c_i < p$, then $\{x_p\}_p = \sum_{i<0} c_i p^i$. It is the unique element of $\mathbb{Z}[1/p] \cap [0,1)$ with $x_p - \{x_p\}_p \in \mathbb{Z}_p$, and it is 0 when $x_p \in \mathbb{Z}_p$. Given $x \in \mathbb{A}_f$, only finitely many $\{x_p\}_p$ are nonzero, so $q = \sum_p \{x_p\}_p$ is a rational number. For each p, the terms $\{x_\ell\}_\ell$ with $\ell \neq p$ lie in $\mathbb{Z}_p$, so $x_p - q \in \mathbb{Z}_p$. Hence $x - q \in \widehat{\mathbb{Z}}$. A rational number lying in $\mathbb{Z}_p$ for every p has no denominator, so it is an integer.

(2) By (1), $(\mathbb{R} \times \widehat{\mathbb{Z}}) \cap \mathbb{Q} = \mathbb{Z}$, and $\mathbb{Z} \cap \big((-\tfrac{1}{2}, \tfrac{1}{2}) \times \widehat{\mathbb{Z}}\big) = \{0\}$, so $\mathbb{Q}$ is discrete. Given $(x_\infty, x_f) \in \mathbb{A}$, write $x_f = q + u$ with $q \in \mathbb{Q}$ and $u \in \widehat{\mathbb{Z}}$ by (1); then $(x_\infty, x_f) = q + (x_\infty - q,\, u)$. □

The standard character. Define $\psi_\infty(x) = \mathbf{e}(x)$ on $\mathbb{R}$ and $\psi_p(x) = \mathbf{e}(-\{x\}_p)$ on $\mathbb{Q}_p$. Each ψ_p is a character of $\mathbb{Q}_p$, because $\{x+y\}_p - \{x\}_p - \{y\}_p \in \mathbb{Z}[1/p] \cap \mathbb{Z}_p = \mathbb{Z}$, and

$$\psi_p(x\mathbb{Z}_p) = 1 \iff x \in \mathbb{Z}_p, \tag{11.1}$$

because if $x \notin \mathbb{Z}_p$ then $x\mathbb{Z}_p$ contains an element with principal part $1/p$. Since ψ_p is trivial on $\mathbb{Z}_p$, the product $\psi = \prod_v \psi_v$ is a well-defined continuous character of $\mathbb{A}$; we write $\psi_f = \prod_p \psi_p$ for its finite part. It is trivial on $\mathbb{Q}$: for $q \in \mathbb{Q}$, the number $q - \sum_p \{q\}_p$ is rational and lies in every $\mathbb{Z}_p$, hence is an integer, so $\psi(q) = \mathbf{e}\big(q - \sum_p \{q\}_p\big) = 1$. Moreover, every character of the compact group $\mathbb{Q}\backslash\mathbb{A}$ is of the form $x \mapsto \psi(\alpha x)$ for a unique $\alpha \in \mathbb{Q}$ [Tat67], [RV99, Ch. 5]. For $\alpha \in \mathbb{Q}$ write $\psi^\alpha(x) = \psi(\alpha x)$.

11.2. **The adelic Heisenberg group and strong approximation.** For any commutative ring A, let $H(A)$ be the set $A^n \times A^n \times A$ with the multiplication (3.1). Then $H(\mathbb{A}) = H(\mathbb{R}) \times H(\mathbb{A}_f)$, where $H(\mathbb{A}_f)$ is the restricted product of the groups $H(\mathbb{Q}_p)$ with respect to the compact open subgroups $H(\mathbb{Z}_p)$. Put

$$K = H(\widehat{\mathbb{Z}}) = \prod_p H(\mathbb{Z}_p),$$

a compact open subgroup of $H(\mathbb{A}_f)$. The group $H(\mathbb{Q})$ embeds diagonally in $H(\mathbb{A})$, and $H(\mathbb{Z}) = \Gamma$ embeds diagonally in $H(\mathbb{R}) \times K$. By Lemma 11.1(2), applied coordinatewise, $H(\mathbb{Q})$ is a discrete subgroup of $H(\mathbb{A})$.

Proposition 11.2 (Strong approximation)**.** (1) $H(\mathbb{A}_f) = H(\mathbb{Q})\,K$*; consequently* $H(\mathbb{A}) = H(\mathbb{Q})\,\big(H(\mathbb{R}) \times K\big)$.
(2) $H(\mathbb{Q}) \cap \big(H(\mathbb{R}) \times K\big) = \Gamma$.

Proof. (1) Let $h_f = (x, y, z) \in H(\mathbb{A}_f)$. By Lemma 11.1(1), choose $a, b \in \mathbb{Q}^n$ with $x - a,\ y - b \in \widehat{\mathbb{Z}}^n$. Since $(a, b, 0)^{-1} = (-a, -b, a \cdot b)$,

$$(a, b, 0)^{-1} h_f = (x - a,\ y - b,\ z + a \cdot b - a \cdot y).$$

Choose $c \in \mathbb{Q}$ with $z + a \cdot b - a \cdot y - c \in \widehat{\mathbb{Z}}$, again by Lemma 11.1(1). Since $(0, 0, c)$ is central, $\gamma = (a, b, 0)(0, 0, c) \in H(\mathbb{Q})$ satisfies $\gamma^{-1} h_f \in K$. For the second statement, apply this to the finite component of $h \in H(\mathbb{A})$; then $\gamma^{-1} h$ has finite component in K and arbitrary real component.

(2) An element of $H(\mathbb{Q})$ whose p-component lies in $H(\mathbb{Z}_p)$ for every p has all its coordinates in $\mathbb{Q} \cap \widehat{\mathbb{Z}} = \mathbb{Z}$. □

Corollary 11.3. *The map $H(\mathbb{R}) \to H(\mathbb{A})$, $g \mapsto (g, 1)$, induces a homeomorphism*

$$\Gamma\backslash H(\mathbb{R}) \xrightarrow{\ \sim\ } H(\mathbb{Q})\backslash H(\mathbb{A})/K$$

that is equivariant for right translation by $H(\mathbb{R})$. Moreover $D \times K$ is a fundamental domain for $H(\mathbb{Q})$ in $H(\mathbb{A})$, where $D = [0, 1)^{2n+1}$.

Proof. Surjectivity is Proposition 11.2(1). If $(g, 1) = \gamma(g', 1)k$ with $\gamma \in H(\mathbb{Q})$ and $k \in K$, the finite components give $\gamma = k^{-1} \in H(\mathbb{Q}) \cap K$, so $\gamma \in \Gamma$ by Proposition 11.2(2), and the real components give $g = \gamma g'$. This proves injectivity, and equivariance is clear.

For the fundamental domain, let $h \in H(\mathbb{A})$. By Proposition 11.2(1), $h = \gamma(g, k)$ with $\gamma \in H(\mathbb{Q})$, $g \in H(\mathbb{R})$ and $k \in K$. By Proposition 4.2, $g = \gamma' d$ with $\gamma' \in \Gamma$ and $d \in D$; since $\gamma' \in K$, we get $h = \gamma\gamma'\,(d, \gamma'^{-1}k)$ with $(d, \gamma'^{-1}k) \in D \times K$. If $\gamma(d, k) = (d', k')$ with both points in $D \times K$, then $\gamma \in H(\mathbb{Q}) \cap K$, so $\gamma \in \Gamma$, and $\gamma d = d'$ forces $\gamma = 1$ by Proposition 4.2. □

Normalize Haar measure on $H(\mathbb{A}_f)$ so that K has volume 1, and on $H(\mathbb{A})$ take the product with dh on $H(\mathbb{R})$. Then $H(\mathbb{Q})\backslash H(\mathbb{A})$ has volume 1, and $H(\mathbb{A})$ acts on $L^2(H(\mathbb{Q})\backslash H(\mathbb{A}))$ by right translation R, as in Proposition 4.4.

Proposition 11.4. *Restriction $\Phi \mapsto F$, $F(g) = \Phi(g,1)$, is a unitary isomorphism of $H(\mathbb{R})$-representations*

$$L^2\big(H(\mathbb{Q})\backslash H(\mathbb{A})\big)^K \xrightarrow{\ \sim\ } L^2(\Gamma\backslash H(\mathbb{R})),$$

where the left side is the space of vectors fixed by $R(K)$.

Proof. If Φ is left $H(\mathbb{Q})$-invariant and right K-invariant, then F is left Γ-invariant, since $(\gamma g, 1) = \gamma \cdot (g,1) \cdot (1, \gamma^{-1})$ in $H(\mathbb{A})$, where γ is embedded diagonally and $(1,\gamma^{-1}) \in K$. By Corollary 11.3, Φ is determined by F, and every Γ-invariant F arises this way. Integrating over the fundamental domain $D \times K$ gives $\|\Phi\|^2 = \int_D |F|^2 \cdot \mathrm{vol}(K) = \|F\|^2$. Equivariance for $H(\mathbb{R})$ is clear. □

In words: *the real quotient is the adelic quotient with the finite places divided out by the maximal compact subgroup K.* The same argument, with strong approximation for SL_2 in place of Proposition 11.2, identifies $L^2(SL_2(\mathbb{Z})\backslash SL_2(\mathbb{R}))$ with the $SL_2(\widehat{\mathbb{Z}})$-fixed vectors in $L^2(SL_2(\mathbb{Q})\backslash SL_2(\mathbb{A}))$; see [Bum97, Ch. 3].

11.3. **Central characters.** The center of $H(\mathbb{A})$ is $Z(\mathbb{A}) \cong \mathbb{A}$, and $Z(\mathbb{Q}) = \mathbb{Q}$. Exactly as in Proposition 6.2, Fourier analysis on the compact group $\mathbb{Q}\backslash\mathbb{A}$ decomposes

$$L^2\big(H(\mathbb{Q})\backslash H(\mathbb{A})\big) = \widehat{\bigoplus}_{\alpha\in\mathbb{Q}} \mathcal{H}^{\mathbb{A}}_\alpha,$$

where $Z(\mathbb{A})$ acts on $\mathcal{H}^{\mathbb{A}}_\alpha$ by ψ^α. Now K contains $Z(\widehat{\mathbb{Z}}) = \widehat{\mathbb{Z}}$, so a nonzero K-fixed vector can lie in $\mathcal{H}^{\mathbb{A}}_\alpha$ only if ψ^α is trivial on $\widehat{\mathbb{Z}}$, which by (11.1) means $\alpha \in \mathbb{Z}$. For $\alpha = m \in \mathbb{Z}$, the restriction of ψ^m to the real center is $s \mapsto \mathbf{e}(ms) = \psi_m(s)$. So under Proposition 11.4,

$$(\mathcal{H}^{\mathbb{A}}_m)^K \cong \mathcal{H}_m \qquad (m \in \mathbb{Z}).$$

The integrality of m, which in Section 6 came from triviality on $\Gamma \cap Z = \mathbb{Z}$, here comes from triviality on $\widehat{\mathbb{Z}}$.

11.4. **The adelic Schrödinger representation.** Let $\mathcal{S}(\mathbb{A}^n) = \mathcal{S}(\mathbb{R}^n) \otimes \mathcal{S}(\mathbb{A}_f^n)$ be the Schwartz–Bruhat space, where $\mathcal{S}(\mathbb{A}_f^n)$ consists of the locally constant, compactly supported functions on $\mathbb{A}_f^n$. Give $\mathbb{A}_f^n$ the Haar measure for which $\widehat{\mathbb{Z}}^n$ has volume 1. For $\alpha \in \mathbb{Q}^\times$, define on $L^2(\mathbb{A}^n)$

$$\big(\rho^{\mathbb{A}}_\alpha(a,b,c)f\big)(t) = \psi\big(\alpha(c + t\cdot b)\big)\, f(t+a) \qquad \big((a,b,c) \in H(\mathbb{A}),\ t \in \mathbb{A}^n\big). \tag{11.2}$$

The computation in Proposition 5.2 shows this is a unitary representation of $H(\mathbb{A})$ with central character ψ^α. Because $\psi = \psi_\infty \psi_f$ and $L^2(\mathbb{A}^n) = L^2(\mathbb{R}^n) \mathbin{\widehat{\otimes}} L^2(\mathbb{A}_f^n)$, it factors as

$$\rho^{\mathbb{A}}_\alpha = \rho_\alpha \otimes \rho^f_\alpha,$$

where ρ_α is the real Schrödinger representation (5.1) (this is where the choice $\psi_\infty(x) = \mathbf{e}(x)$ matters) and ρ^f_α is given by the same formula on $L^2(\mathbb{A}_f^n)$ with ψ_f in place of ψ.

Theorem 11.5 (Weil)**.** *Let $\alpha \in \mathbb{Q}^\times$.*

(1) *$\rho^{\mathbb{A}}_\alpha$ is irreducible.*

(2) *For $f \in \mathcal{S}(\mathbb{A}^n)$ and $h \in H(\mathbb{A})$, the adelic theta series*

$$\Theta_f^{\mathbb{A}}(h) = \sum_{k \in \mathbb{Q}^n} \big(\rho_\alpha^{\mathbb{A}}(h) f\big)(k)$$

converges absolutely, is left $H(\mathbb{Q})$-invariant, and satisfies $R(g)\Theta_f^{\mathbb{A}} = \Theta^{\mathbb{A}}_{\rho_\alpha^{\mathbb{A}}(g)f}$. The map $f \mapsto \Theta_f^{\mathbb{A}}$ extends to a unitary isomorphism of $\rho_\alpha^{\mathbb{A}}$ onto $\mathcal{H}_\alpha^{\mathbb{A}}$.

In particular, $\rho_\alpha^{\mathbb{A}}$ occurs in $L^2(H(\mathbb{Q})\backslash H(\mathbb{A}))$ with multiplicity exactly one.

Comments on the proof. The invariance in (2) is the computation of Proposition 9.1(1), now with no cosets needed: for $\gamma = (a, b, c) \in H(\mathbb{Q})$ and $k \in \mathbb{Q}^n$, the number $\alpha(c + k \cdot b)$ is rational, so the phase $\psi(\alpha(c + k \cdot b))$ is 1 because ψ is trivial on $\mathbb{Q}$, and translation by $a \in \mathbb{Q}^n$ permutes $\mathbb{Q}^n$. Equivariance is formal, as in Proposition 9.1(2). For the irreducibility of $\rho_\alpha^{\mathbb{A}}$ and the fact that the theta map is unitary and surjective, see [Wei64], [Mum91, Ch. I], and [Fox89]. The proof of surjectivity is the argument of Section 8, with the compact group $\mathbb{Z}^n\backslash\mathbb{R}^n$ replaced by $\mathbb{Q}^n\backslash\mathbb{A}^n$, whose character group is $\mathbb{Q}^n$ via ψ. The heart of that argument was the identity (8.3): translation by a in the lattice shifts Fourier indices by ma. Adelically, translation by $a \in \mathbb{Q}^n$ shifts the Fourier index $k \in \mathbb{Q}^n$ by αa, and since $\alpha \neq 0$ this action of $\mathbb{Q}^n$ on $\mathbb{Q}^n$ is *transitive*. There is one orbit instead of the $|m|^n$ orbits of $m\mathbb{Z}^n$ on $\mathbb{Z}^n$, and this is why the multiplicity drops to one. □

11.5. **Recovering the multiplicity $|m|^n$.** Combining Proposition 11.4, the computation of central characters, and Theorem 11.5, we obtain for $m \neq 0$ isomorphisms of $H(\mathbb{R})$-representations

$$\mathcal{H}_m \;\cong\; (\mathcal{H}_m^{\mathbb{A}})^K \;\cong\; \big(\rho_m \otimes \rho_m^f\big)^K = \rho_m \otimes \big(L^2(\mathbb{A}_f^n)\big)^{\rho_m^f(K)}, \tag{11.3}$$

since K acts only on the second factor. So everything reduces to computing the K-fixed vectors of the finite part.

Proposition 11.6. *Let $m \neq 0$. The space of $f \in L^2(\mathbb{A}_f^n)$ fixed by $\rho_m^f(K)$ has dimension $|m|^n$, with orthonormal basis the indicator functions*

$$\mathbf{1}_{j/m + \widehat{\mathbb{Z}}^n}, \qquad j \in \mathbb{Z}^n/m\mathbb{Z}^n.$$

Proof. Every element of K is a product $(0, b, 0)(a, 0, 0)(0, 0, c)$ with $a, b \in \widehat{\mathbb{Z}}^n$ and $c \in \widehat{\mathbb{Z}}$, so it suffices to impose invariance under these three kinds of elements.

- $(0, 0, c)$ acts by $\psi_f(mc) = 1$, since $mc \in \widehat{\mathbb{Z}}$.
- $(a, 0, 0)$ acts by translation, so invariance means f is $\widehat{\mathbb{Z}}^n$-periodic.
- $(0, b, 0)$ acts by multiplication by $\psi_f(m\, t \cdot b)$. Invariance for all $b \in \widehat{\mathbb{Z}}^n$ means $f(t) = 0$ for almost every t outside $\{t : \psi_f(m\, t \cdot \widehat{\mathbb{Z}}^n) = 1\}$, which by (11.1) is $\{t : mt \in \widehat{\mathbb{Z}}^n\} = \frac{1}{m}\widehat{\mathbb{Z}}^n$. (It suffices to use b in the countable dense subset $\mathbb{Z}^n \subset \widehat{\mathbb{Z}}^n$, since ψ_f is continuous.)

So f is supported on the compact group $\frac{1}{m}\widehat{\mathbb{Z}}^n$ and is invariant under its open subgroup $\widehat{\mathbb{Z}}^n$; that is, f is a function on the finite group $\frac{1}{m}\widehat{\mathbb{Z}}^n/\widehat{\mathbb{Z}}^n \cong (\mathbb{Z}/m\mathbb{Z})^n$. The cosets $j/m + \widehat{\mathbb{Z}}^n$, $j \in \mathbb{Z}^n/m\mathbb{Z}^n$, are disjoint of volume 1, so their indicator functions form an orthonormal basis, and conversely each of them is K-fixed. □

Remark 11.7 (Local factorization). The count is a product of local counts. For a prime p, let $\rho_{m,p}$ be the representation of $H(\mathbb{Q}_p)$ on $L^2(\mathbb{Q}_p^n)$ given by (11.2) with

ψ_p in place of ψ. The argument of Proposition 11.6 shows that at a prime $p \nmid m$ the only $\rho_{m,p}(H(\mathbb{Z}_p))$-fixed vectors are the multiples of $\mathbf{1}_{\mathbb{Z}_p^n}$, and at $p \mid m$ the fixed vectors form a space of dimension $p^{n\,v_p(m)}$, spanned by the indicators of the cosets of $\mathbb{Z}_p^n$ in $p^{-v_p(m)}\mathbb{Z}_p^n$. The product of these dimensions is $\prod_p p^{n\,v_p(m)} = |m|^n$. So the multiplicity $|m|^n$ is concentrated at the primes dividing m, and it comes entirely from the finite places.

Finally, the adelic theta series specializes to the theta series of Section 9.

Theorem 11.8. *For $f \in \mathcal{S}(\mathbb{R}^n)$, $j \in \mathbb{Z}^n$ and $g \in H(\mathbb{R})$,*

$$\Theta^{\mathbb{A}}_{f \otimes \mathbf{1}_{j/m+\widehat{\mathbb{Z}}^n}}(g,1) = \Theta_{f,j}(g).$$

Consequently, under (11.3), the decomposition $\mathcal{H}_m = \widehat{\bigoplus}_r \Theta_r(L^2(\mathbb{R}^n))$ of Theorem 9.2 is ρ_m tensored with the basis of Proposition 11.6, and the multiplicity space $\mathrm{Hom}_H(\rho_m, \mathcal{H}_m)$ of Corollary 9.5 and Remark 9.8 is the space of K-fixed vectors in ρ_m^f.

Proof. The finite component of $(g,1)$ is the identity, so for $k \in \mathbb{Q}^n$

$$\big(\rho_m^{\mathbb{A}}(g,1)(f \otimes \mathbf{1}_{j/m+\widehat{\mathbb{Z}}^n})\big)(k) = \big(\rho_m(g)f\big)(k) \cdot \mathbf{1}_{j/m+\widehat{\mathbb{Z}}^n}(k).$$

The indicator is 1 exactly when $k - j/m \in \mathbb{Q}^n \cap \widehat{\mathbb{Z}}^n = \mathbb{Z}^n$, by Lemma 11.1(1). So the sum over $k \in \mathbb{Q}^n$ becomes the sum over $k \in j/m + \mathbb{Z}^n$, which is (9.1). □

This is the adelic explanation of the shifted lattices $j/m + \mathbb{Z}^n$ that appeared, somewhat mysteriously, in the definition of $\Theta_{f,j}$: they are the rational points of the K-fixed cosets $j/m + \widehat{\mathbb{Z}}^n$. The finite Heisenberg group of Section 9.5 has a transparent adelic meaning as well. Let $\Phi \in (\mathcal{H}_m^{\mathbb{A}})^K$ correspond to $F \in \mathcal{H}_m$, and let $\nu = \nu_{u,v} \in H(\mathbb{Q})$. By left invariance of Φ under the diagonal element (ν, ν),

$$(L_{u,v}F)(g) = \Phi(\nu^{-1}g, 1) = \Phi(g, \nu) = \big(R(1,\nu)\Phi\big)(g,1).$$

So $L_{u,v}$ is right translation by the finite adele $(1,\nu) \in H(\mathbb{A}_f)$. This element does not lie in K, but by (3.3) it normalizes K up to central elements of $\frac{1}{m}\widehat{\mathbb{Z}}$, on which ψ^m is trivial; so it preserves the K-fixed vectors, and under (11.3) it acts by $\rho_m^f(\nu)$. One checks directly that $\rho_m^f(\nu_{u,0})\mathbf{1}_{j/m+\widehat{\mathbb{Z}}^n} = \mathbf{1}_{(j-u)/m+\widehat{\mathbb{Z}}^n}$ and $\rho_m^f(\nu_{0,v})\mathbf{1}_{j/m+\widehat{\mathbb{Z}}^n} = \mathbf{e}(-j\cdot v/m)\,\mathbf{1}_{j/m+\widehat{\mathbb{Z}}^n}$, in agreement with Proposition 9.7(2). For the second identity, note that $\psi_f(q) = \mathbf{e}(-q)$ for $q \in \mathbb{Q}$, because ψ is trivial on $\mathbb{Q}$.

12. Further directions

We close by indicating briefly where the ideas above lead.

12.1. **General nilmanifolds.** For a simply connected nilpotent Lie group G and a lattice Γ, Kirillov's orbit method parametrizes the irreducible unitary representations of G by coadjoint orbits, and the representations occurring in $L^2(\Gamma\backslash G)$ are those attached to "integral" orbits. Their multiplicities are computed by the formulas of Moore and of Howe and Richardson [Moo65, How71, Ric71], and the finer structure of the invariant subspaces was studied by Auslander and Brezin [AB73]; see [CG90, Ch. 5] for an exposition and [Fox89] for an adelic proof. Theorem 9.3 is the first nontrivial case.

12.2. **Jacobi forms and Siegel modular forms.** The Jacobi group is the semidirect product $H_1(\mathbb{R}) \rtimes SL_2(\mathbb{R})$, and its automorphic forms combine the Heisenberg variable of Section 9 with the modular variable of the transformation law in Theorem 10.4. These are the Jacobi forms of Eichler and Zagier [EZ85]; see [BS98] for the representation-theoretic view. Since the Heisenberg group is the unipotent radical of a maximal parabolic subgroup of $Sp_4(\mathbb{R})$, the Fourier–Jacobi coefficients of Siegel modular forms of degree two are Jacobi forms, and their analysis begins with the decomposition proved here.

12.3. **The Weil representation and theta correspondences.** The uniqueness in the Stone–von Neumann theorem makes the Weil representation of the metaplectic group well defined [Wei64]. Restricting it to dual reductive pairs gives Howe's theta correspondence [How80], one of the main sources of automorphic forms on higher-rank groups. See also [LV80].

12.4. **Abelian varieties.** Mumford replaced the real Heisenberg group by finite and adelic Heisenberg groups attached to line bundles on abelian varieties, obtaining an algebraic theory of theta functions valid in all characteristics [Mum66, Mum91]; see also [Pol03]. The multiplicity space of Remark 9.8, which Section 11 identifies with the K-fixed vectors of the finite-adelic Schrödinger representation, is the analytic shadow of that theory.

12.5. **Analysis and dynamics.** The Weil–Brezin map is the Zak transform of time-frequency analysis, where it underlies the theory of Gabor frames [Grö01]. On the dynamical side, equidistribution of flows on $\Gamma\backslash H$ gives estimates for theta sums [FF06], and Lerch zeta functions have a natural home on the same nilmanifold [Lag].

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